

# BFS Traversal

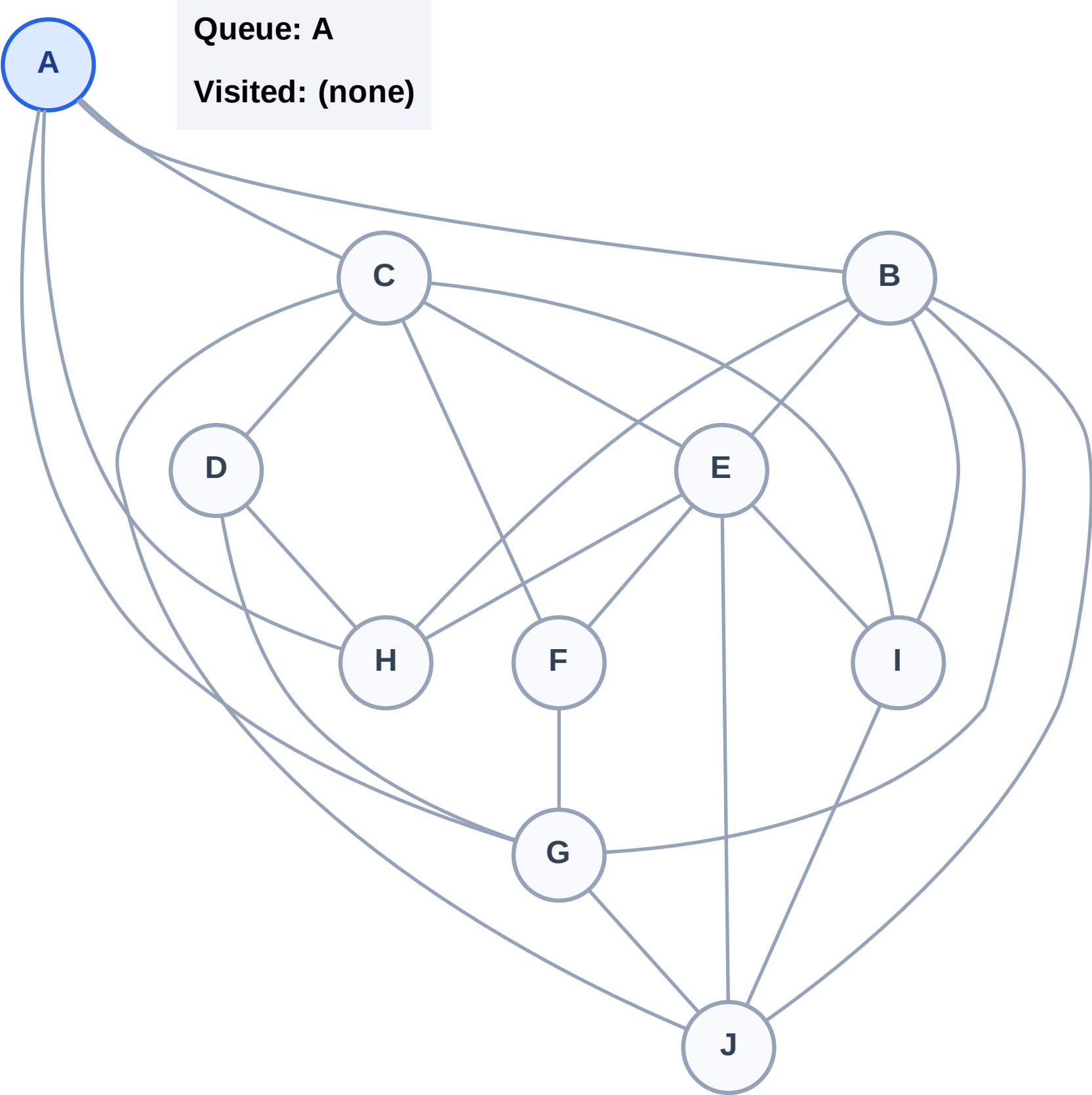
Step 1: Start at A

Complete

Processing

Discovered

Unseen



# BFS Traversal

Step 2: Process node A

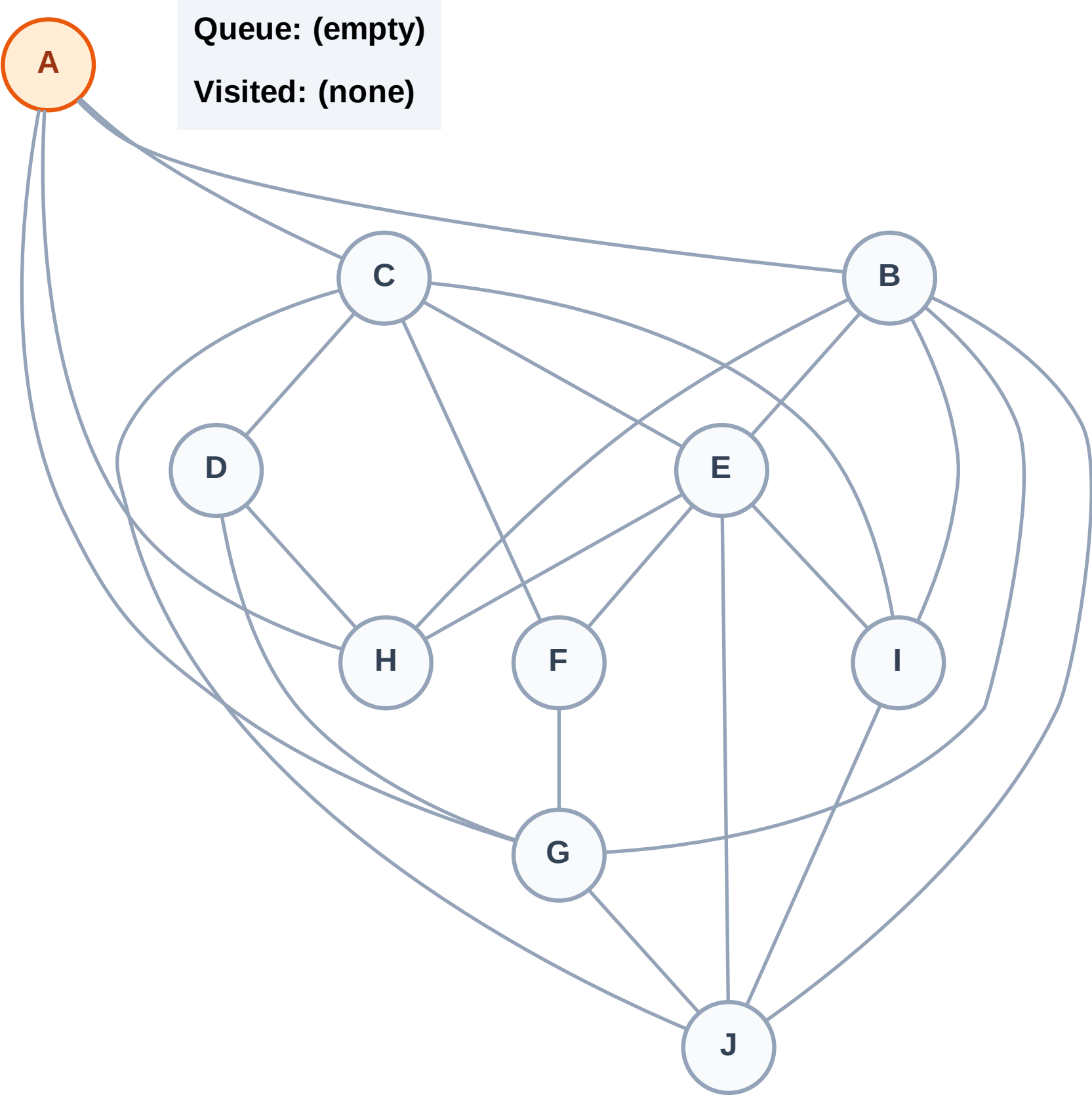
Legend

Complete

Processing

Discovered

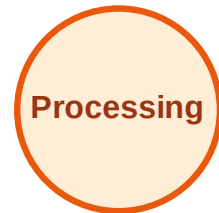
Unseen



# BFS Traversal

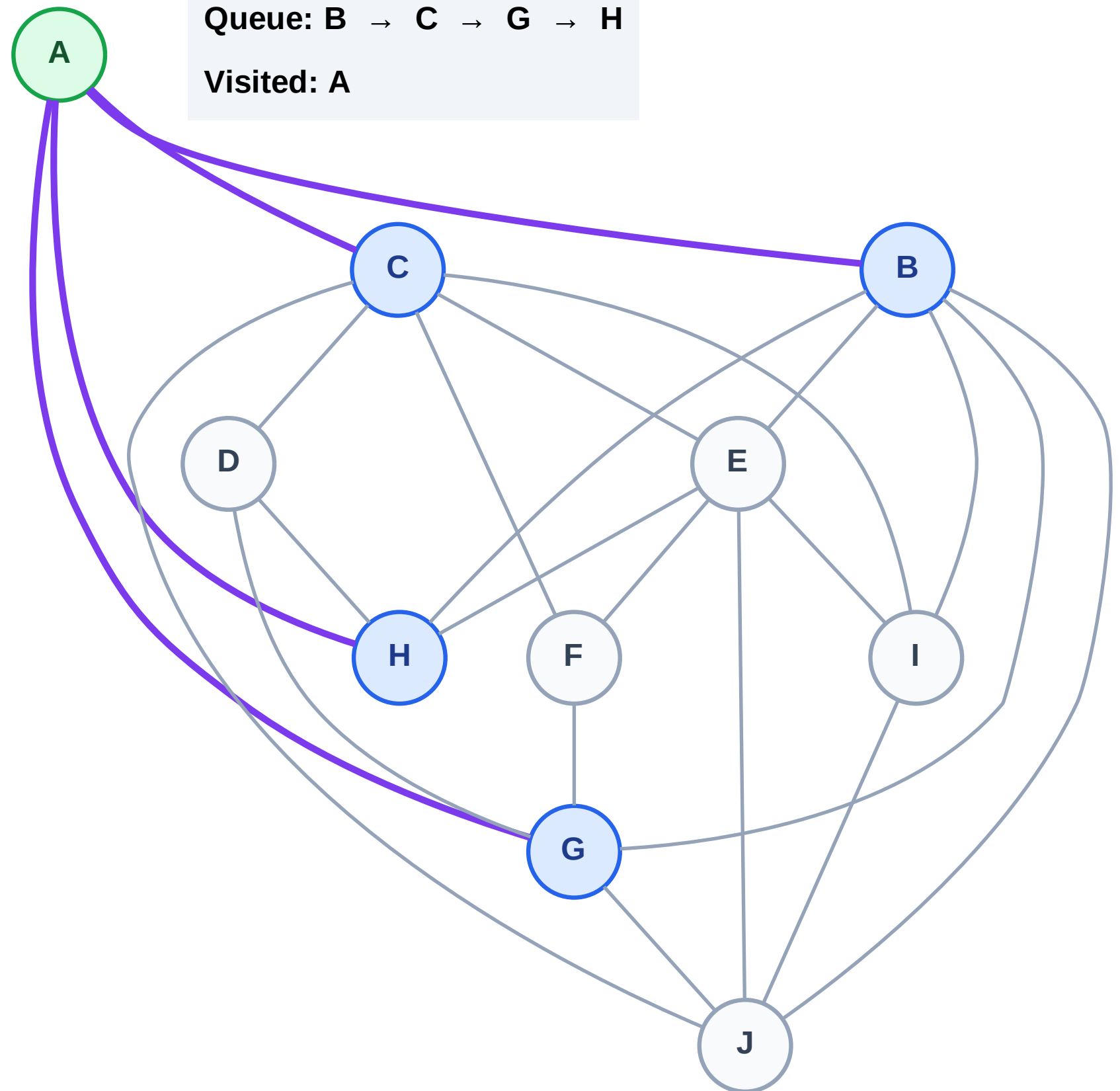
Step 3: Discover B, C, G, H from A

## Legend



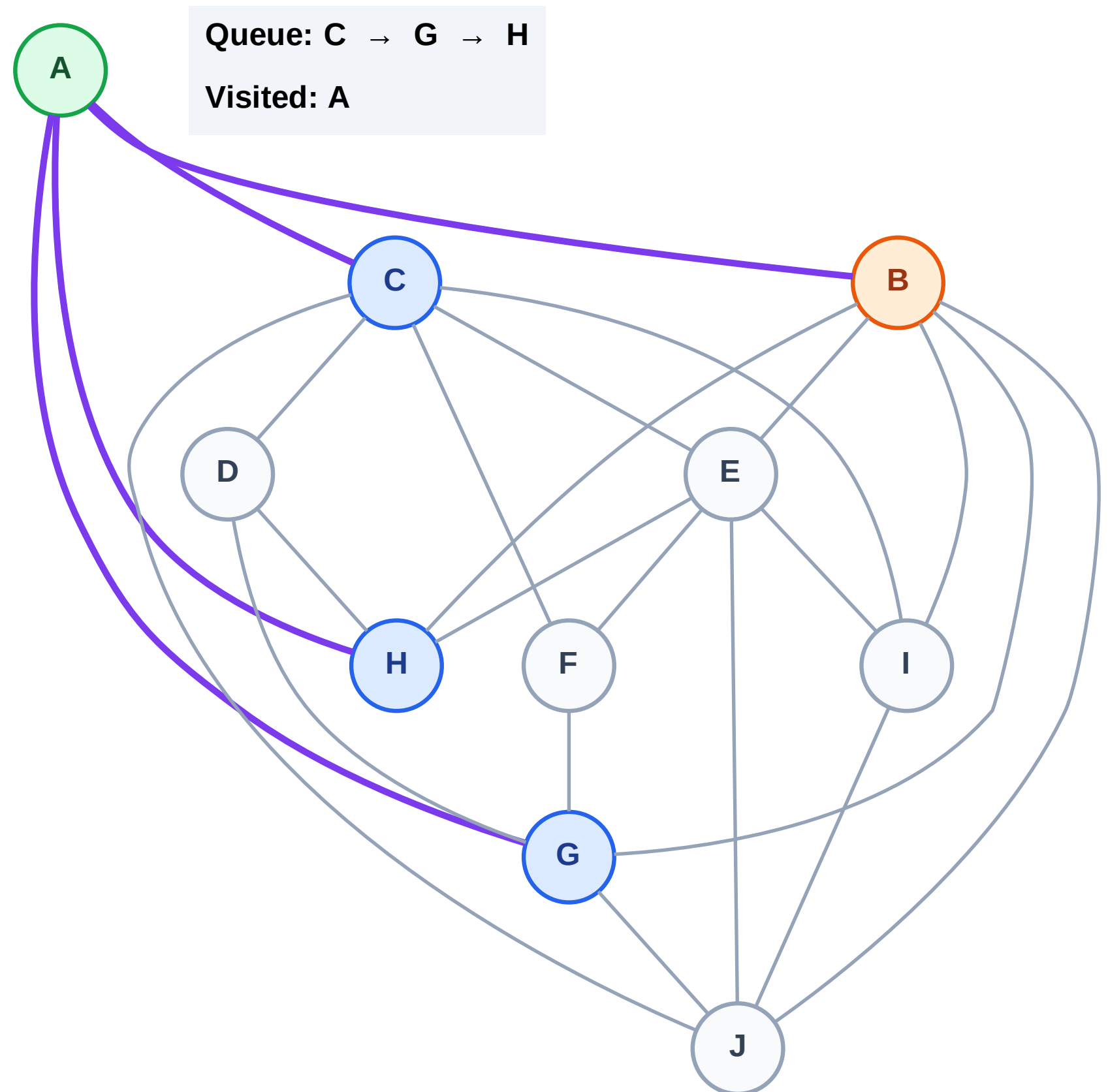
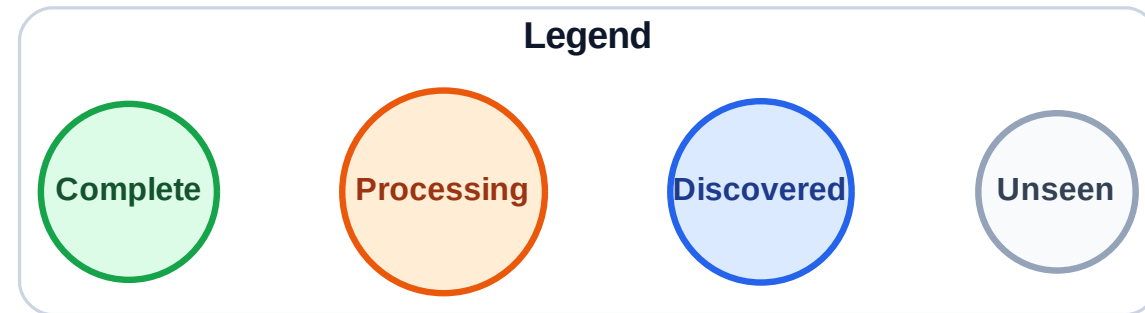
Queue: B → C → G → H

Visited: A



# BFS Traversal

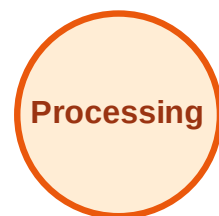
Step 4: Process node B



# BFS Traversal

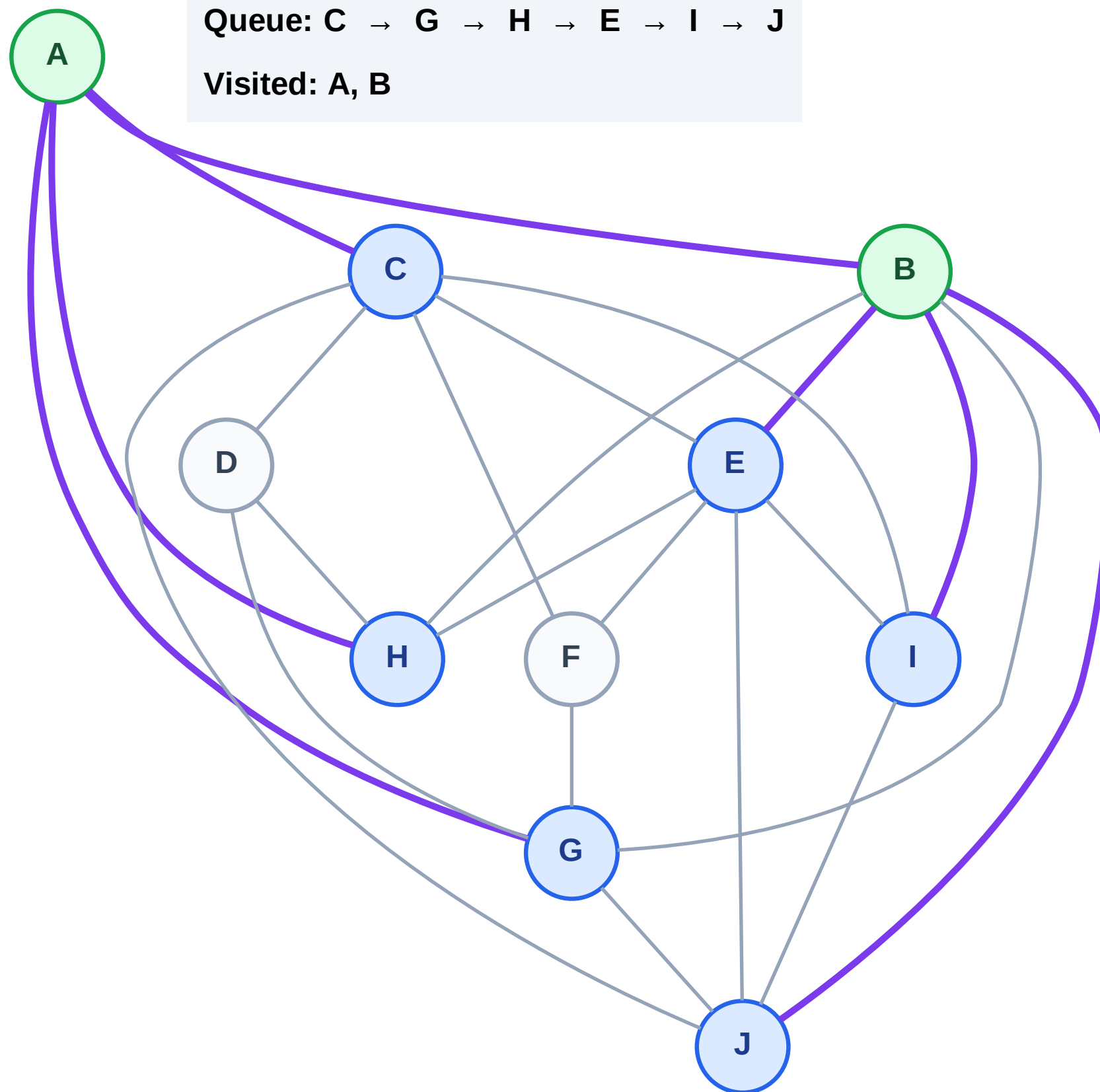
Step 5: Discover E, I, J from B

## Legend



Queue: C → G → H → E → I → J

Visited: A, B



# BFS Traversal

Step 6: Process node C

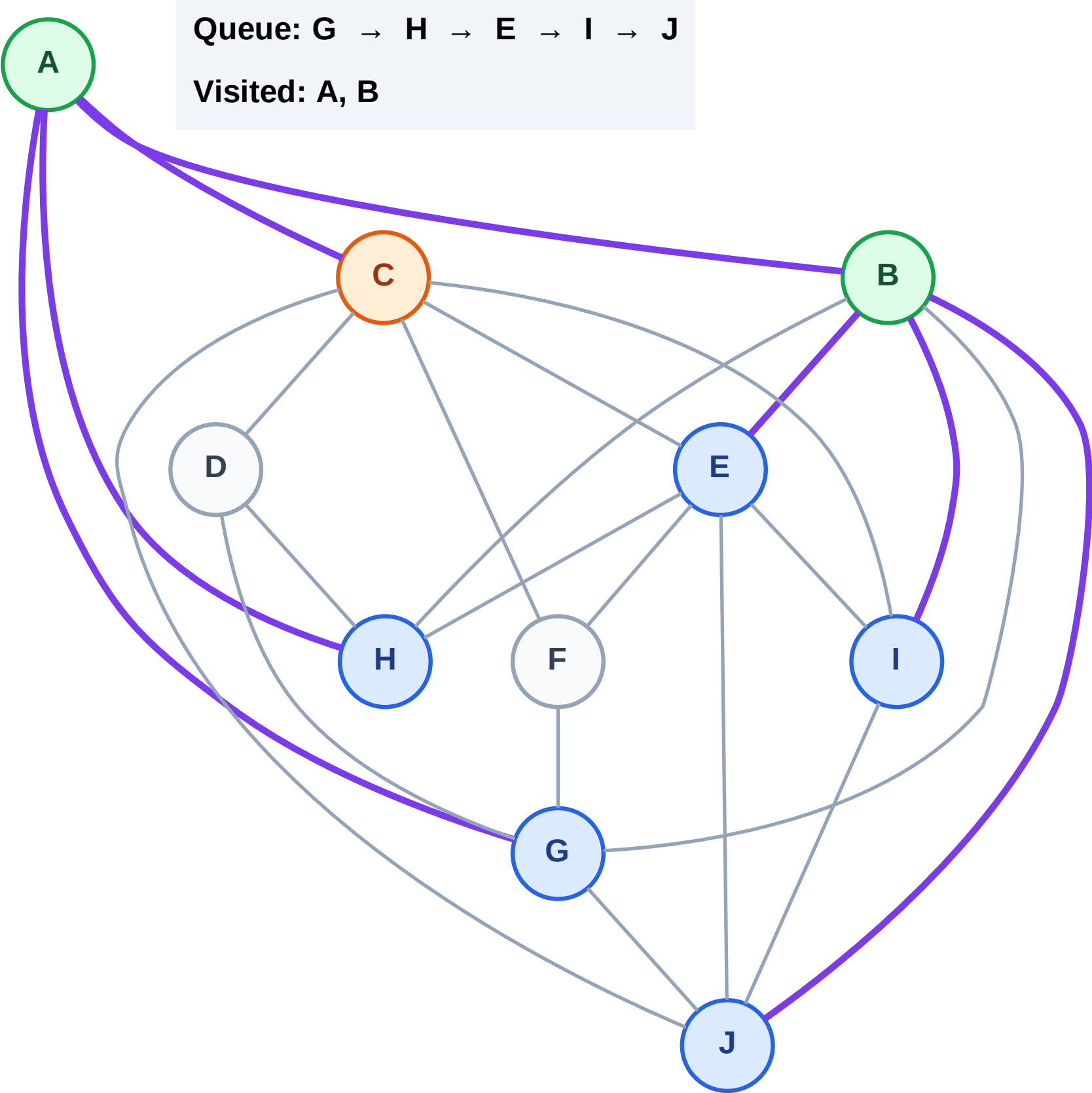
Legend

Complete

Processing

Discovered

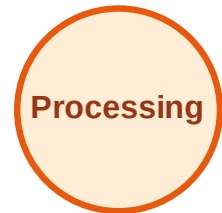
Unseen



# BFS Traversal

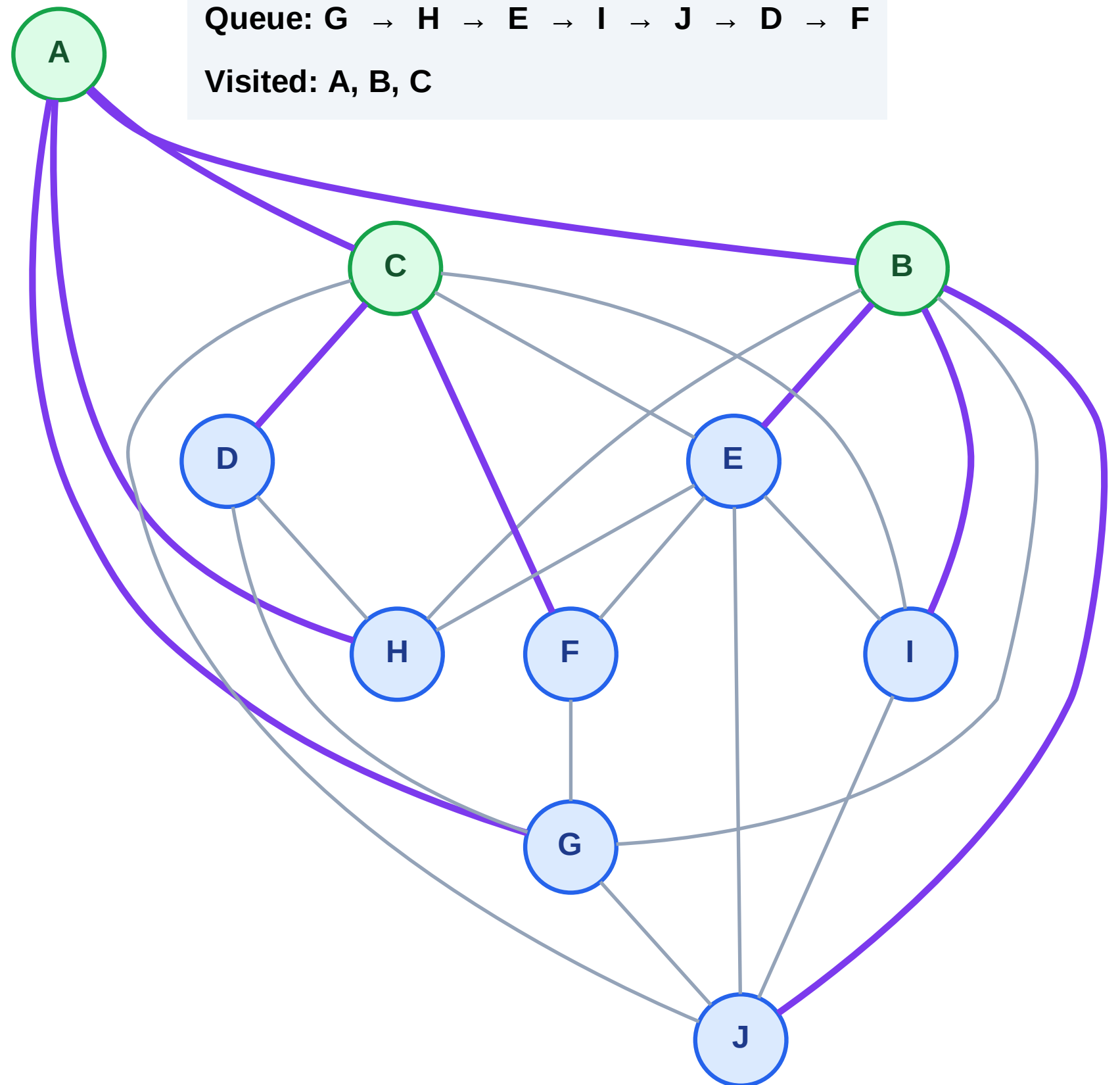
Step 7: Discover D, F from C

## Legend



Queue: G → H → E → I → J → D → F

Visited: A, B, C



# BFS Traversal

Step 8: Process node G

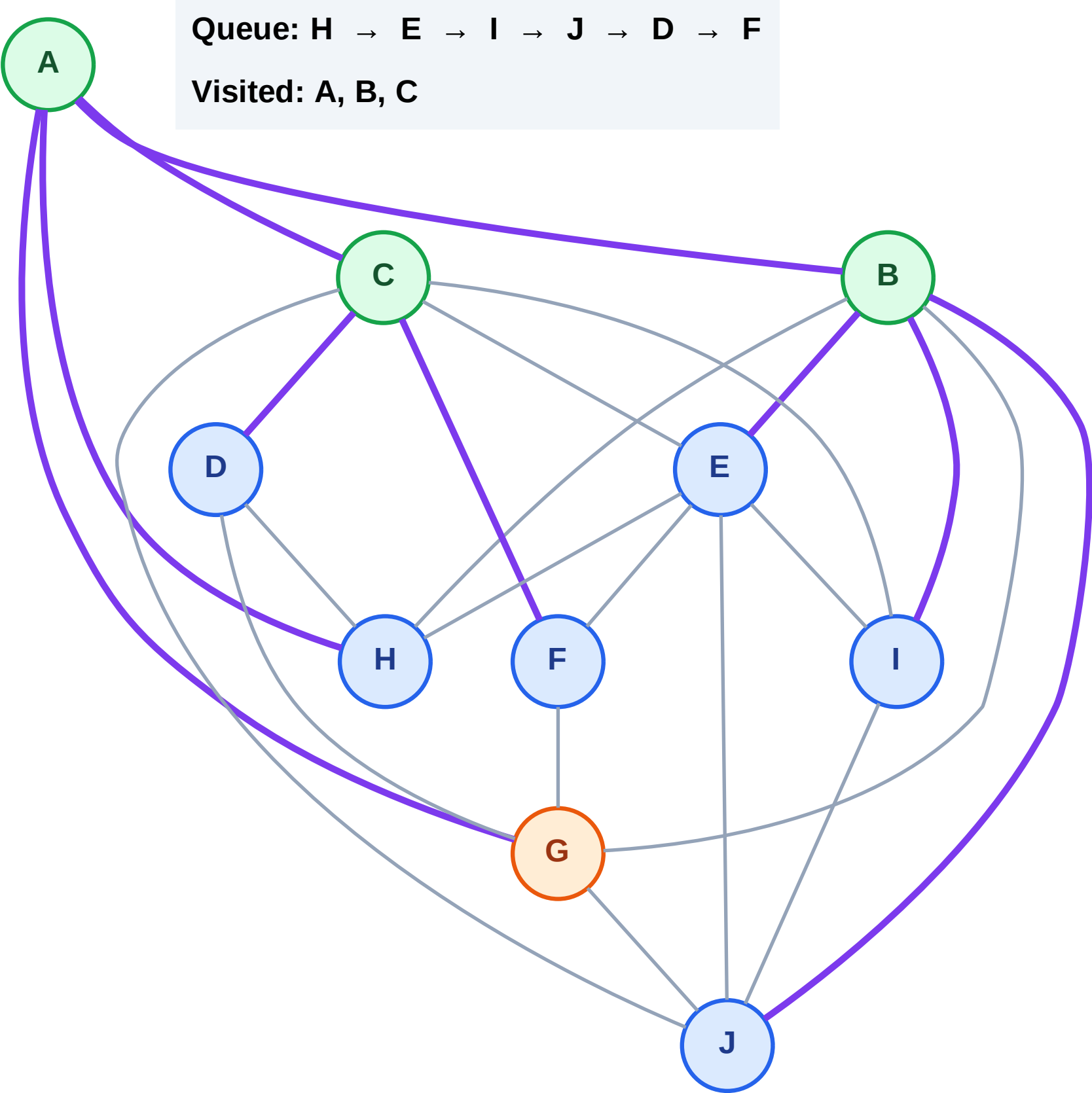
Legend

Complete

Processing

Discovered

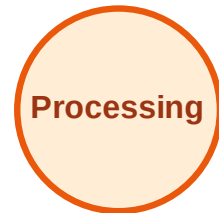
Unseen



# BFS Traversal

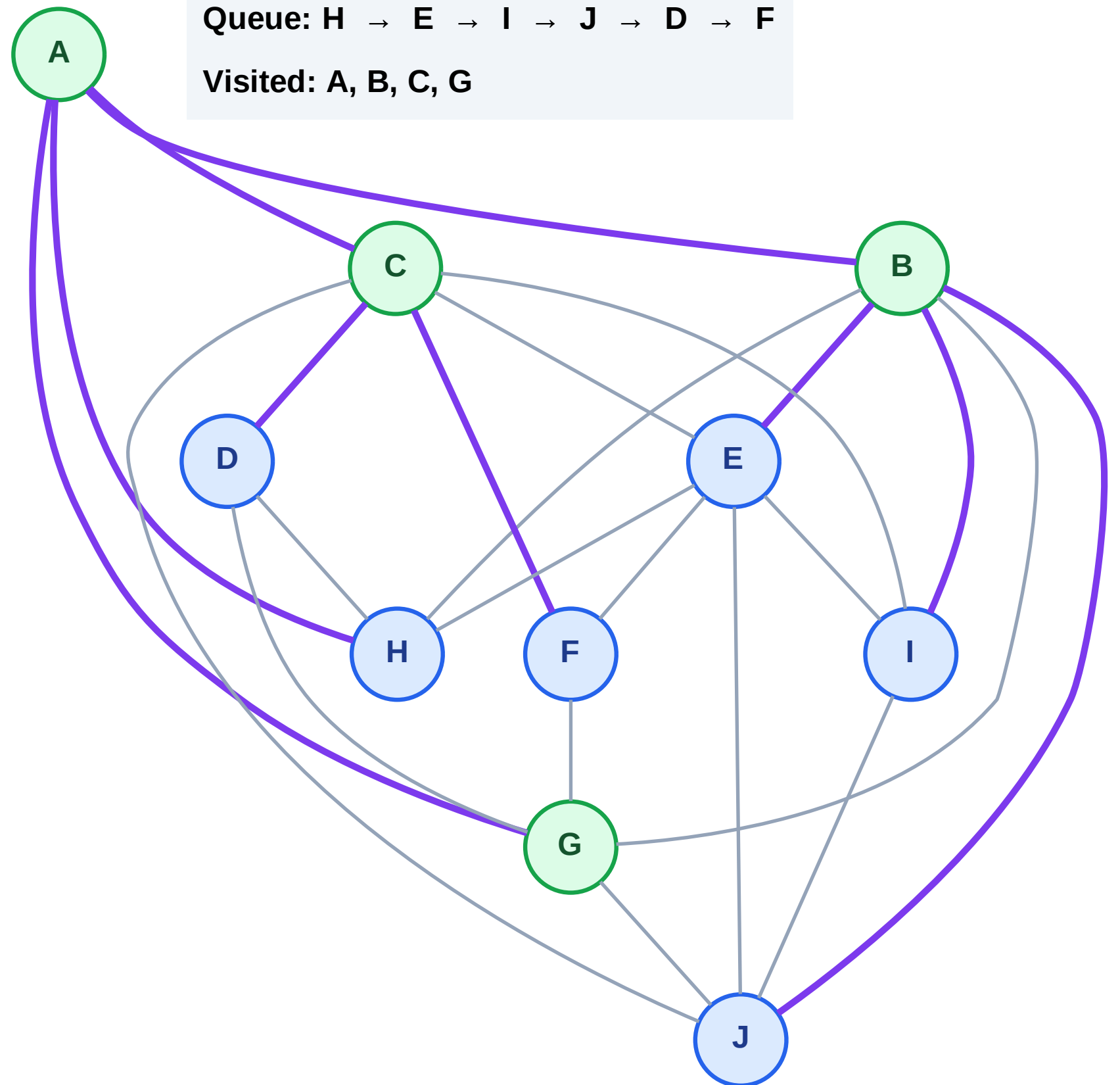
Step 9: No new neighbors from G

## Legend



Queue: H → E → I → J → D → F

Visited: A, B, C, G



# BFS Traversal

Step 10: Process node H

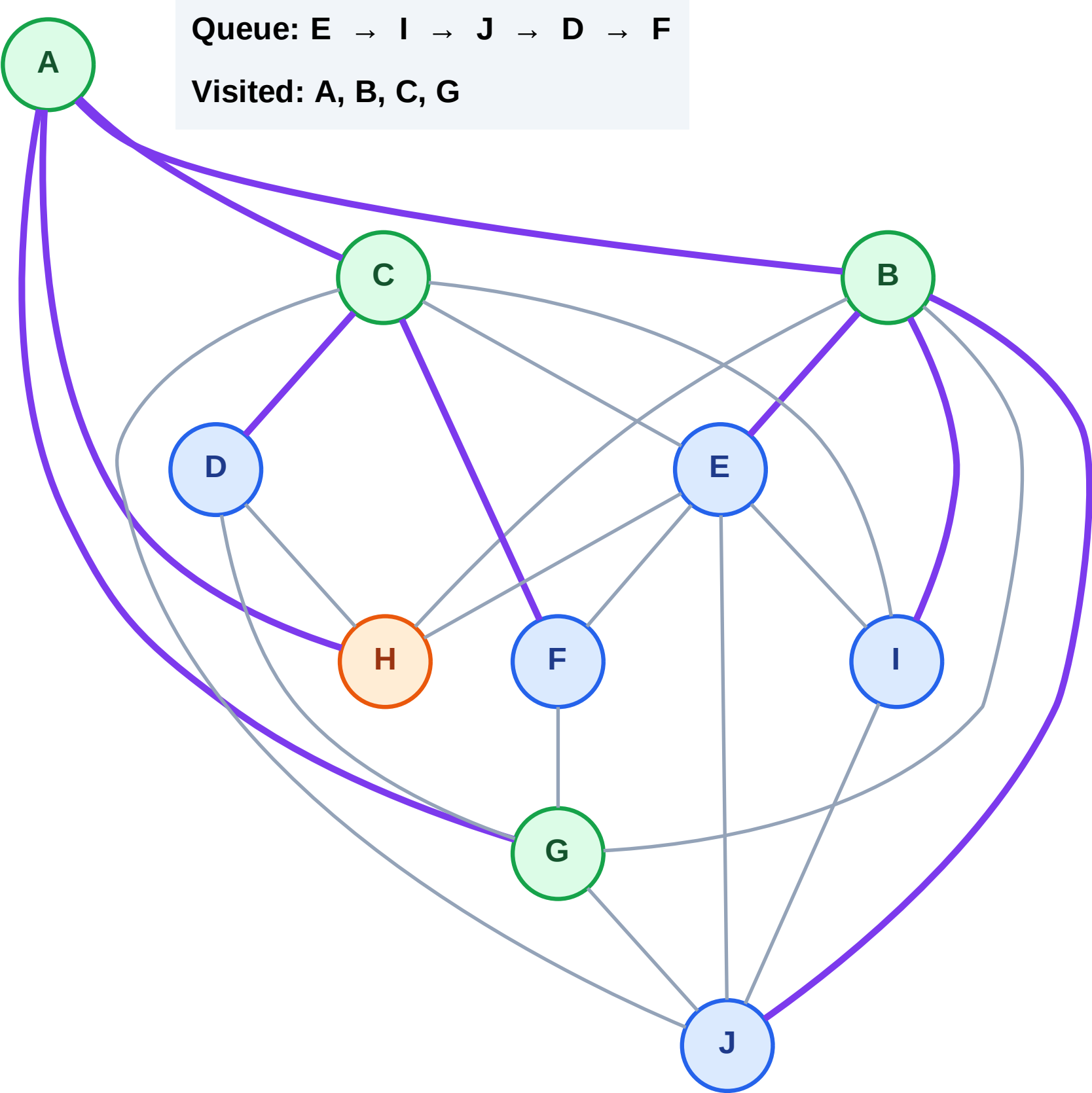
Legend

Complete

Processing

Discovered

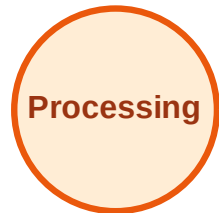
Unseen



# BFS Traversal

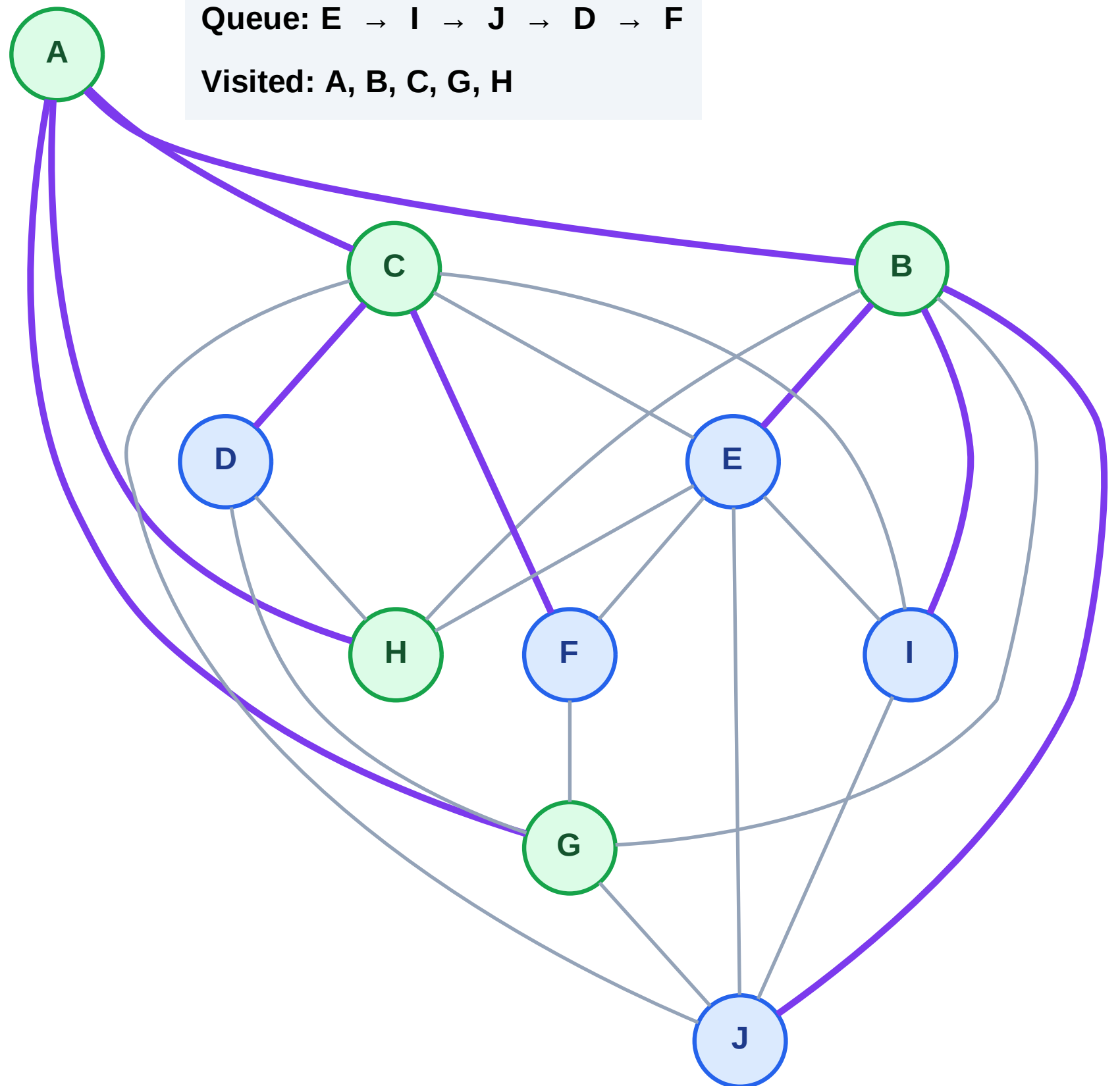
Step 11: No new neighbors from H

## Legend



Queue: E → I → J → D → F

Visited: A, B, C, G, H



# BFS Traversal

Step 12: Process node E

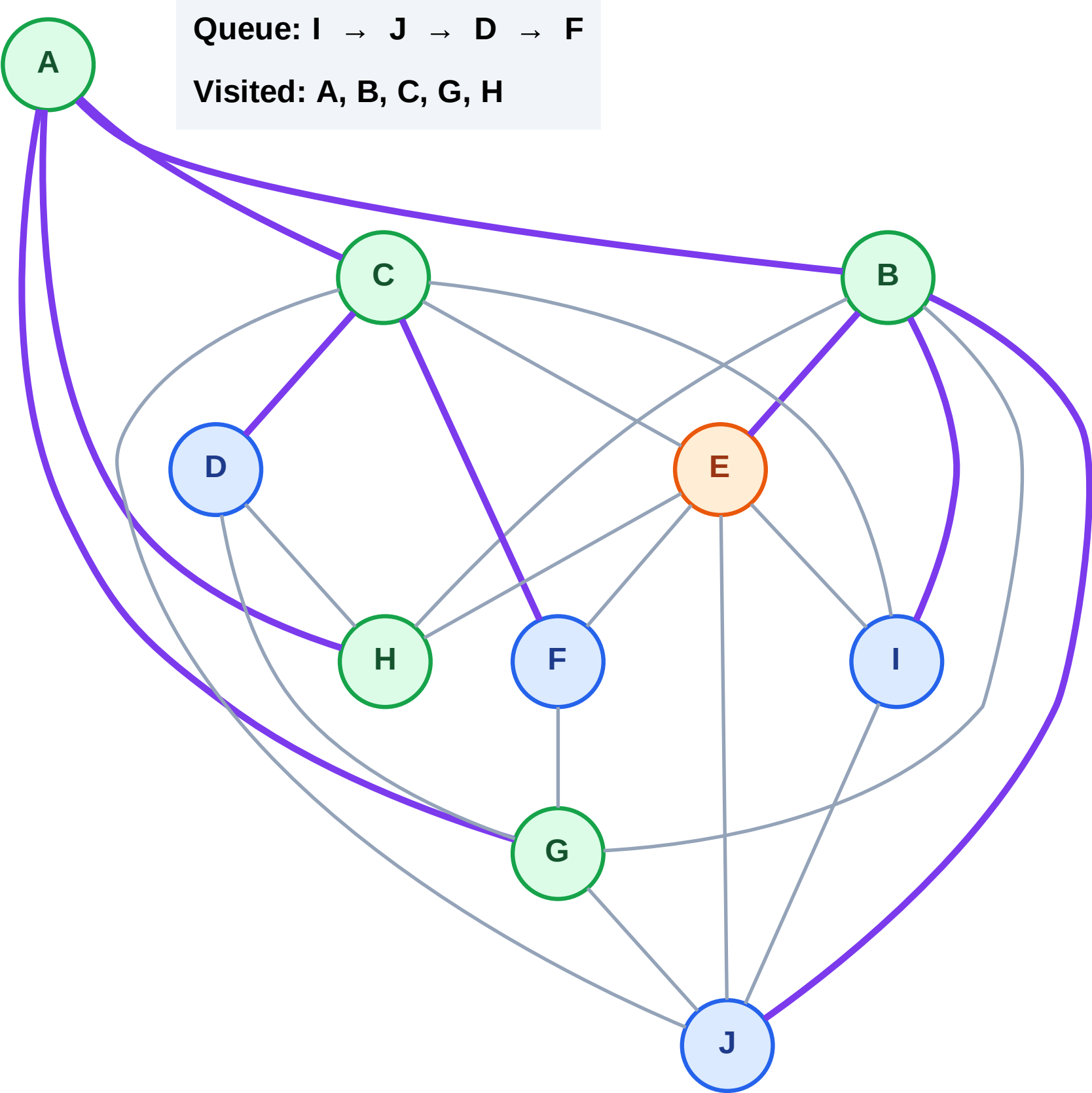
Legend

Complete

Processing

Discovered

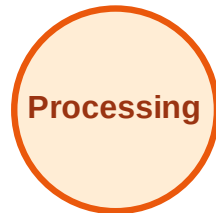
Unseen



# BFS Traversal

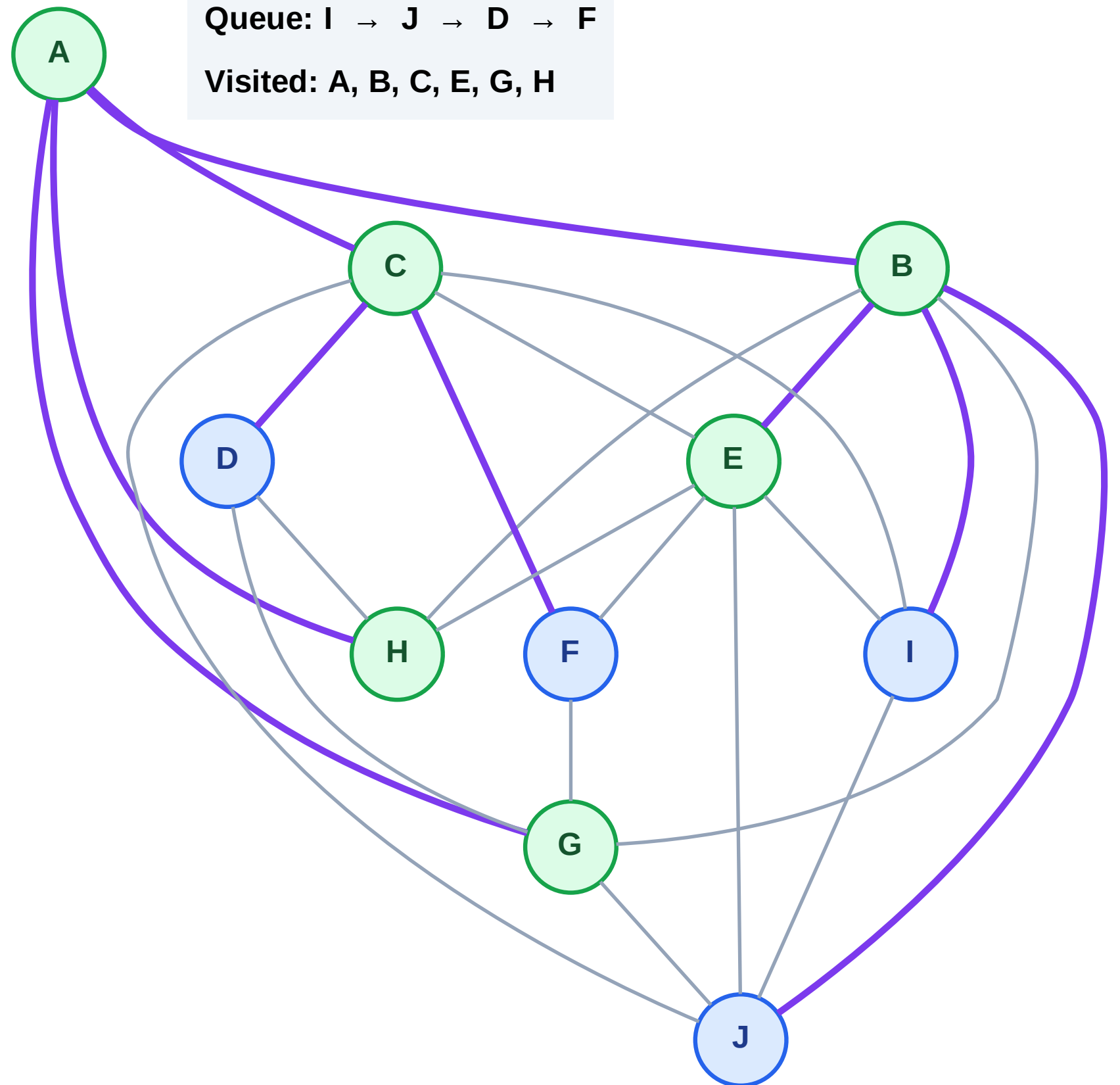
Step 13: No new neighbors from E

## Legend



Queue: I → J → D → F

Visited: A, B, C, E, G, H



# BFS Traversal

Step 14: Process node I

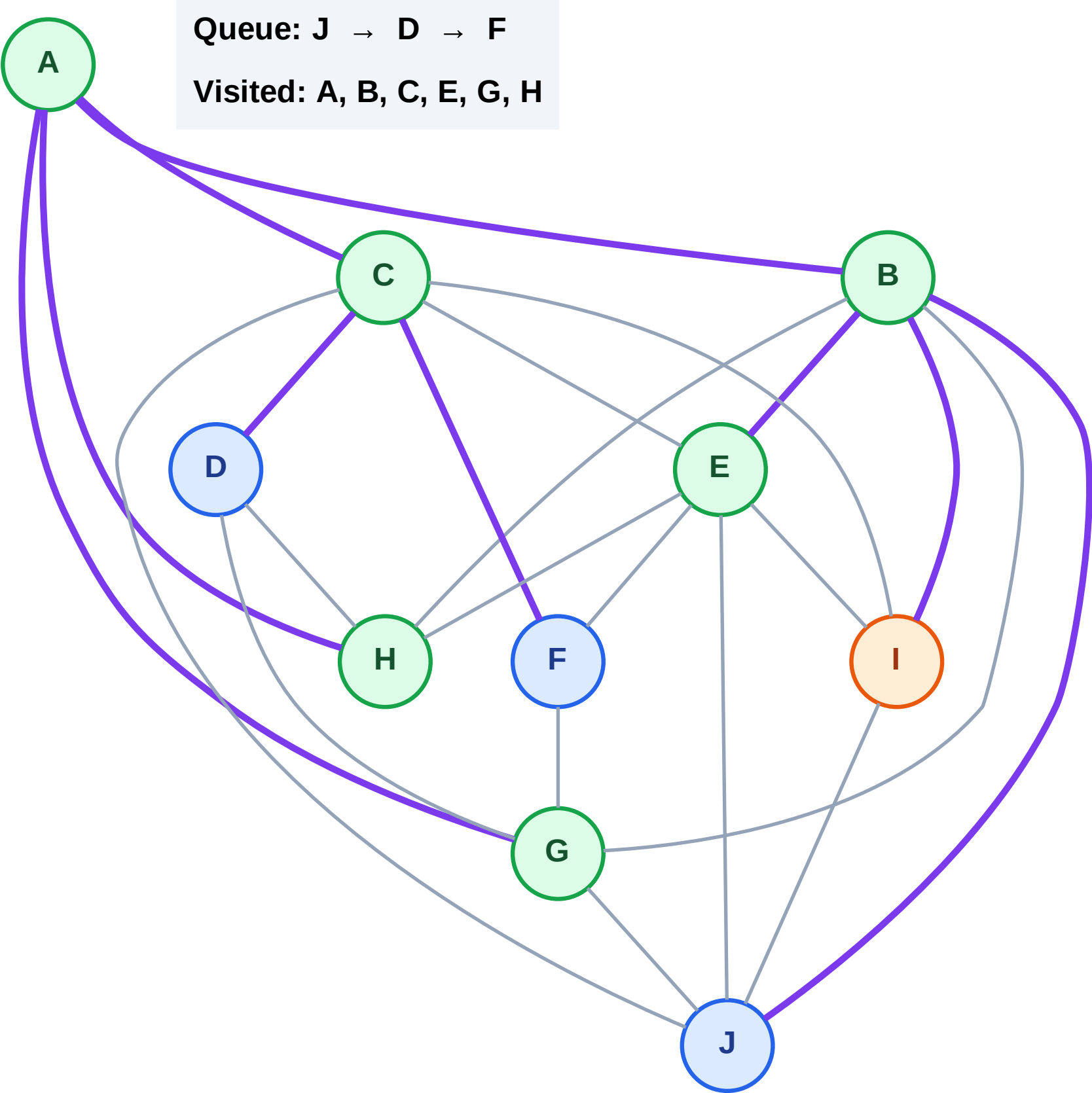
Legend

Complete

Processing

Discovered

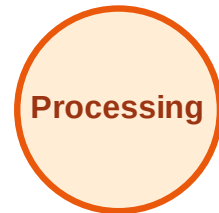
Unseen



# BFS Traversal

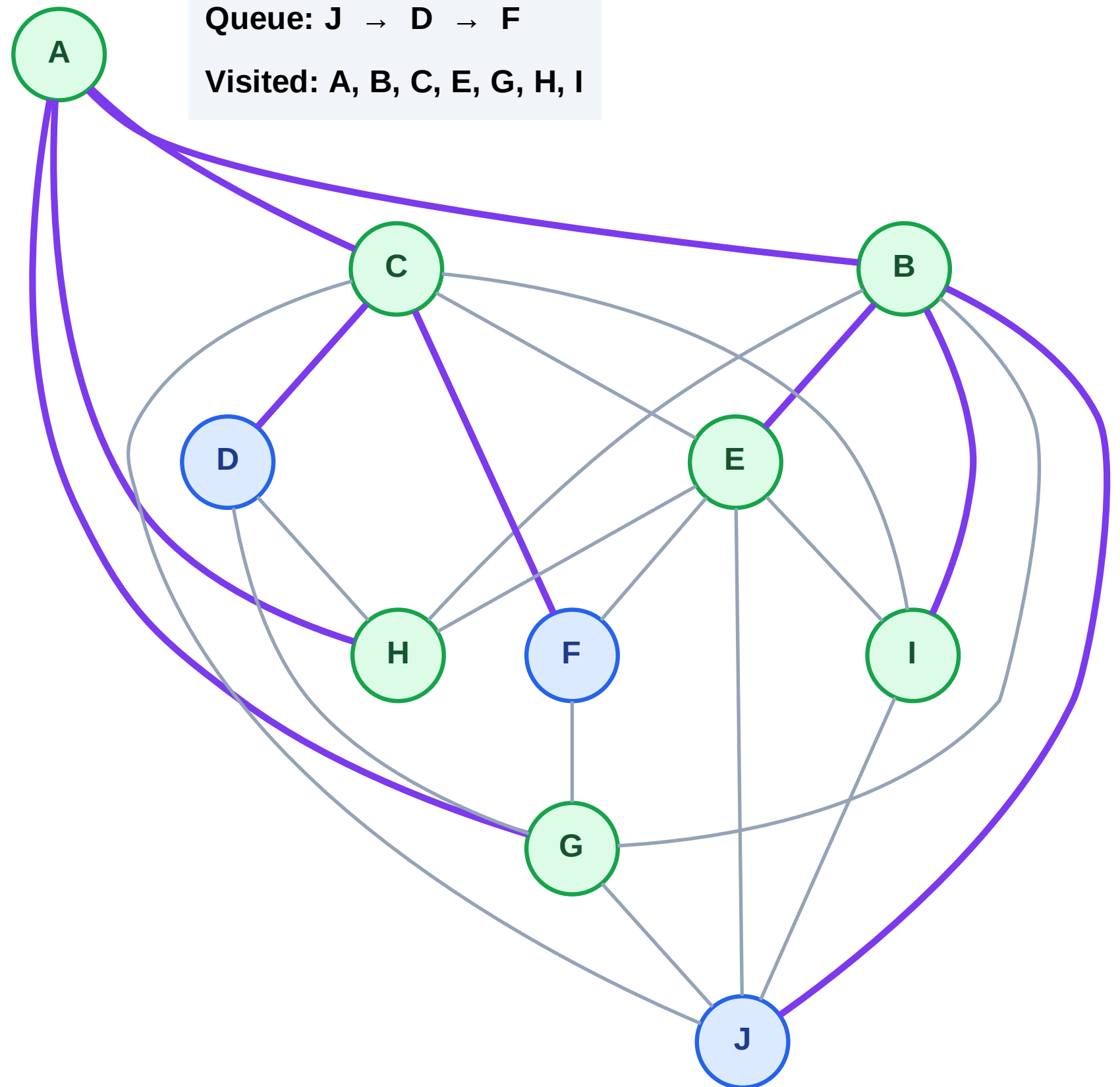
Step 15: No new neighbors from I

## Legend



Queue: J → D → F

Visited: A, B, C, E, G, H, I



# BFS Traversal

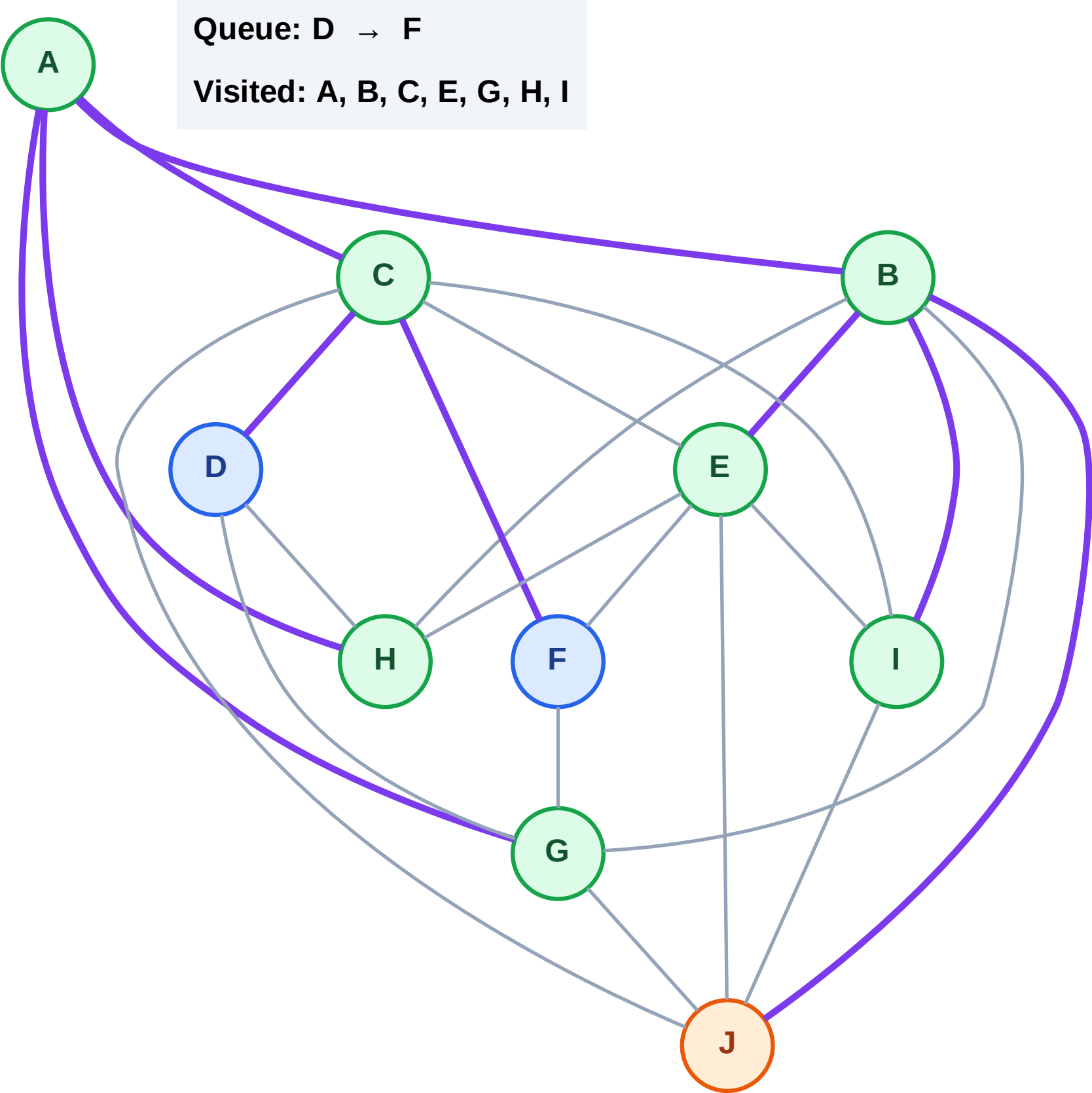
Step 16: Process node J

Complete

Processing

Discovered

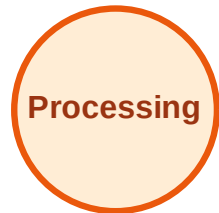
Unseen



# BFS Traversal

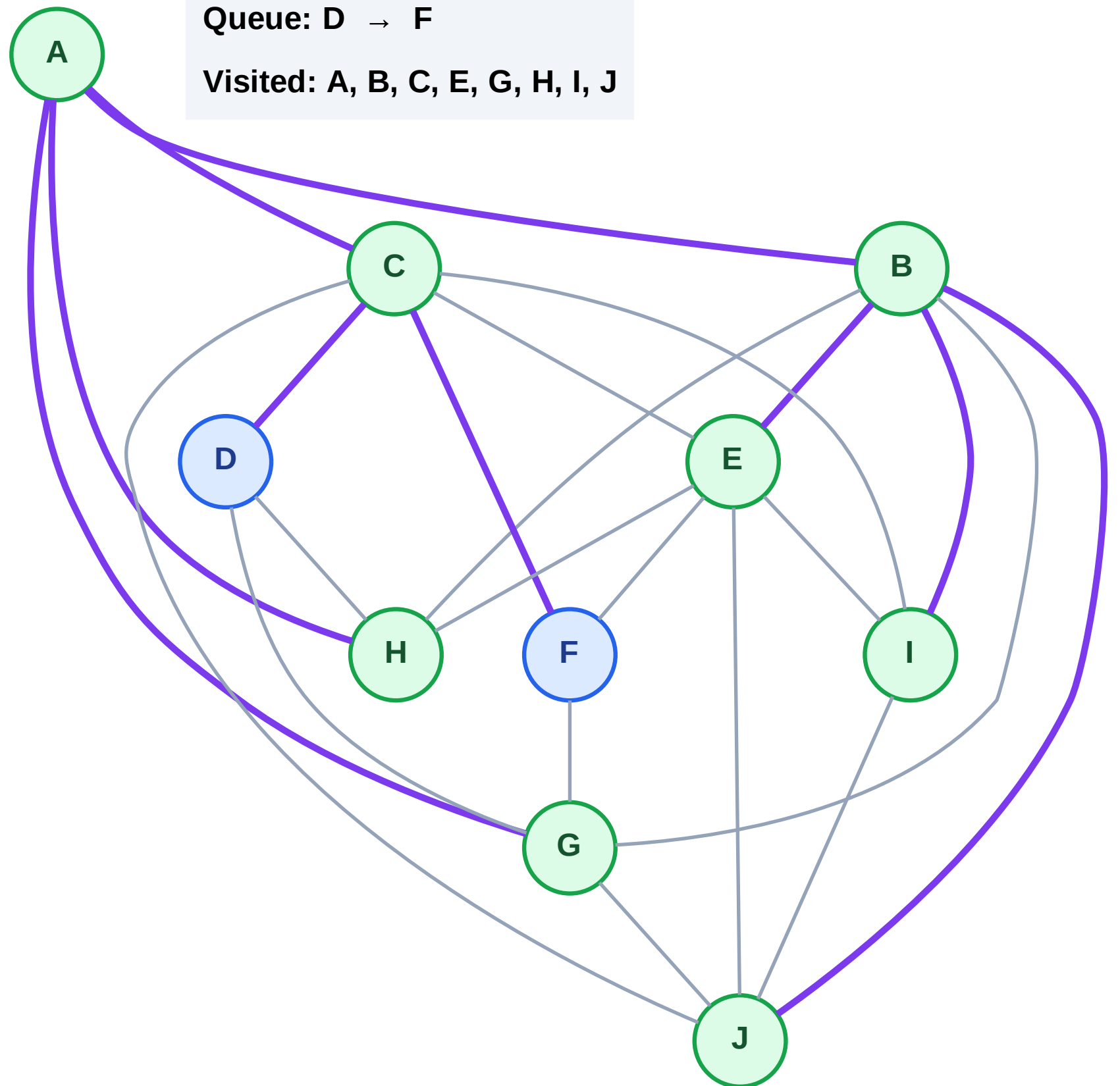
Step 17: No new neighbors from J

## Legend



Queue: D → F

Visited: A, B, C, E, G, H, I, J



# BFS Traversal

Step 18: Process node D

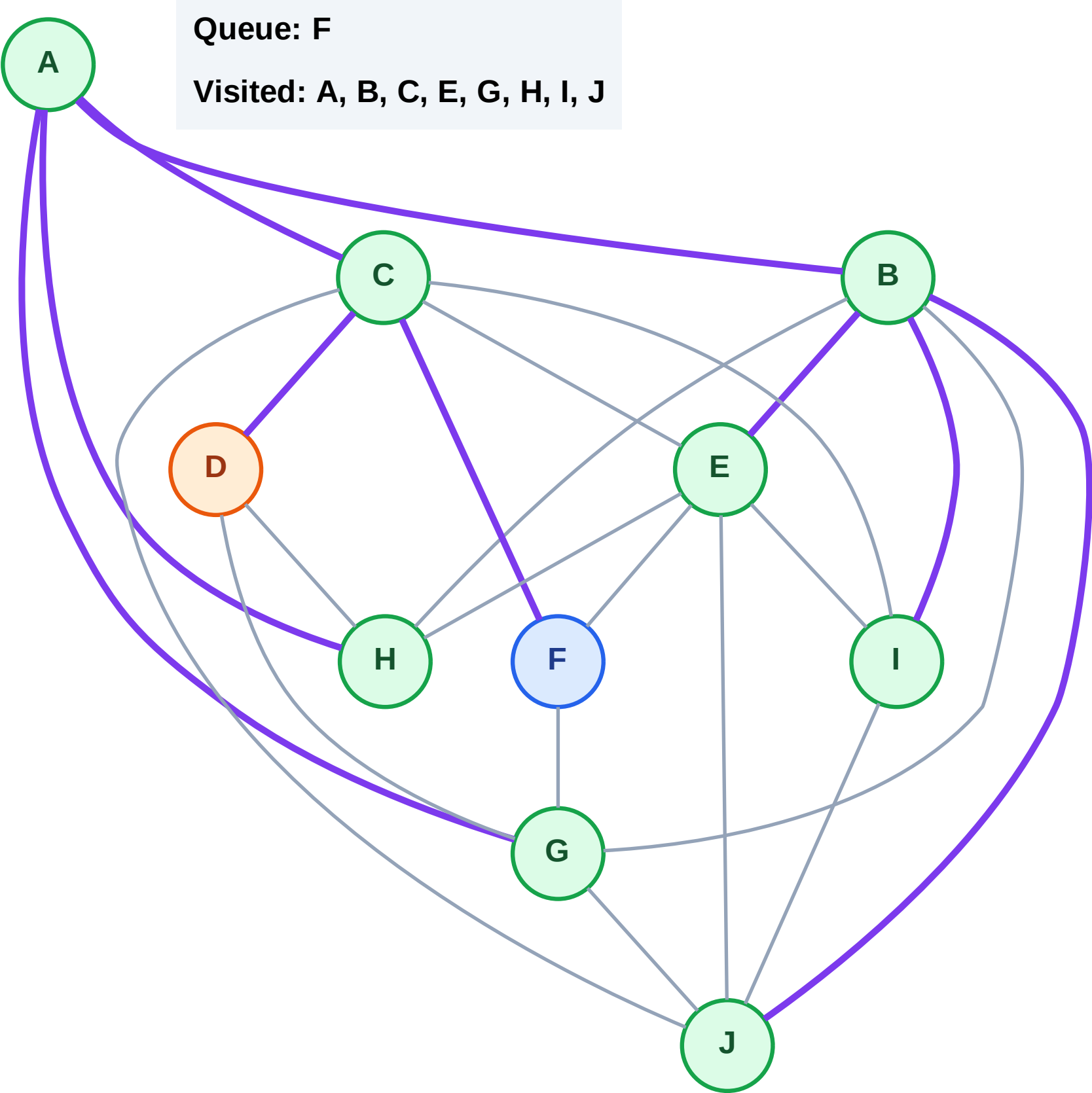
Legend

Complete

Processing

Discovered

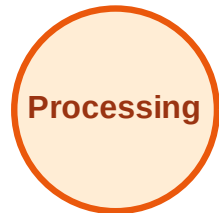
Unseen



# BFS Traversal

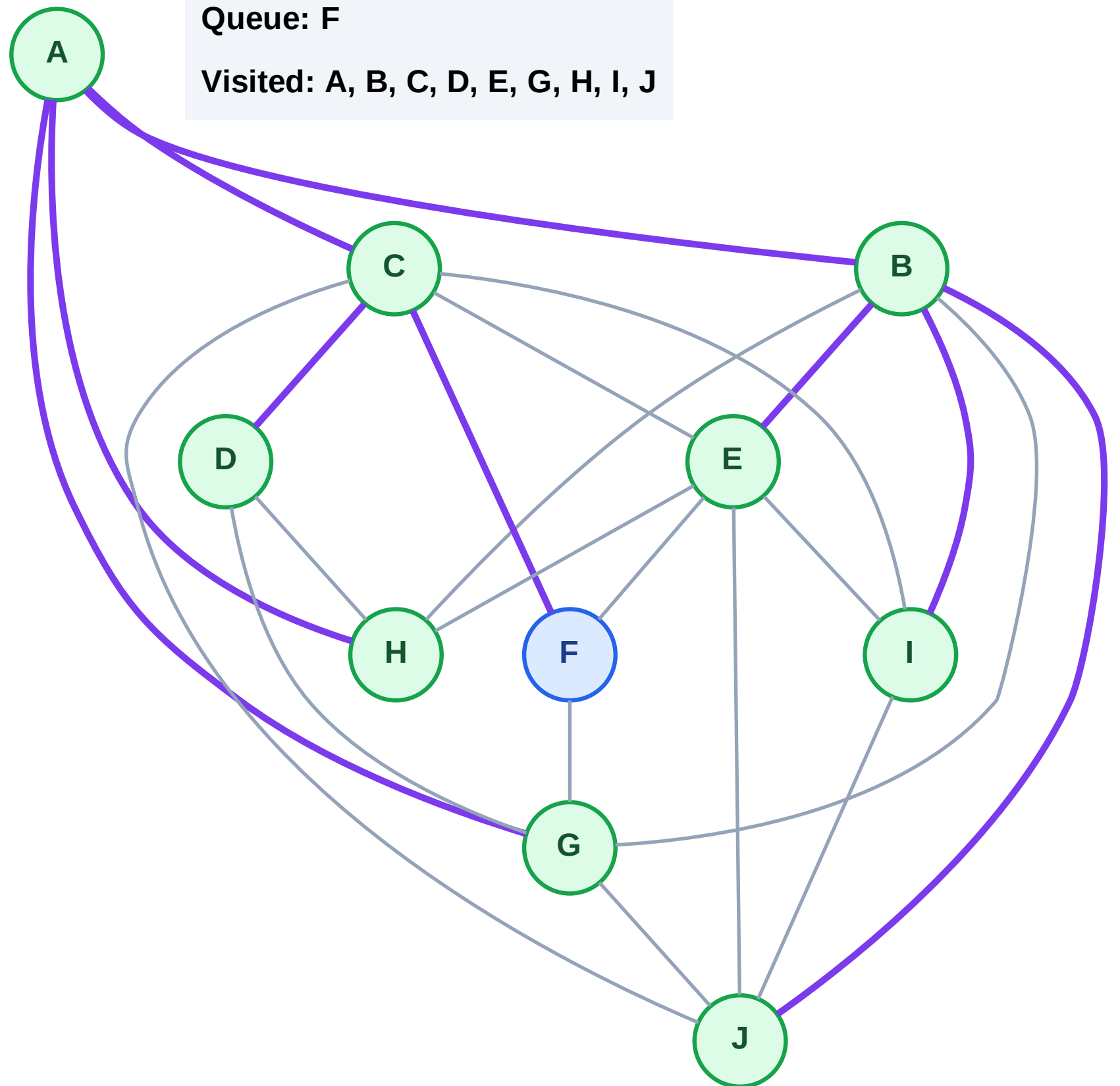
Step 19: No new neighbors from D

## Legend



Queue: F

Visited: A, B, C, D, E, G, H, I, J



# BFS Traversal

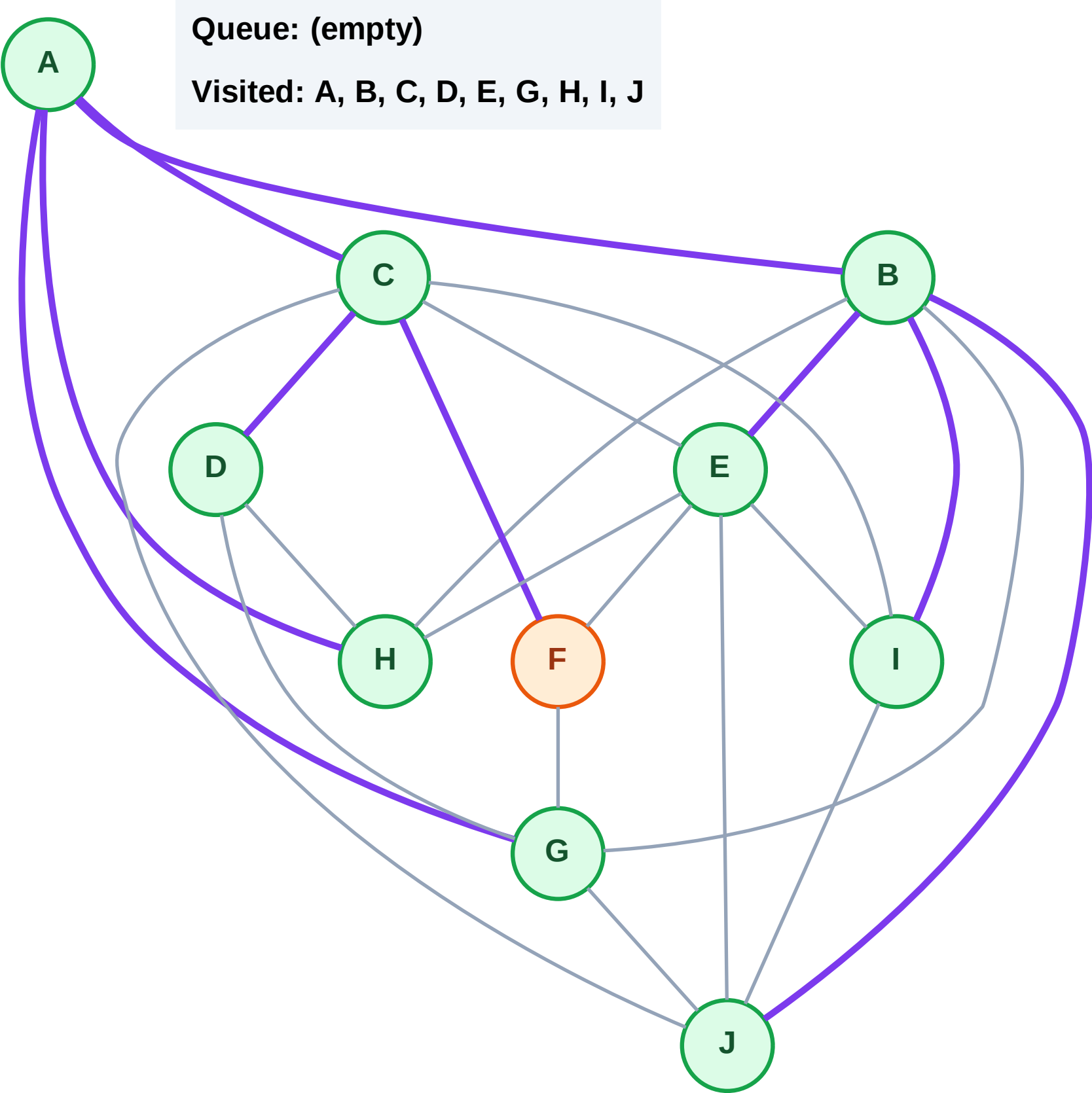
Step 20: Process node F

Complete

Processing

Discovered

Unseen



# BFS Traversal

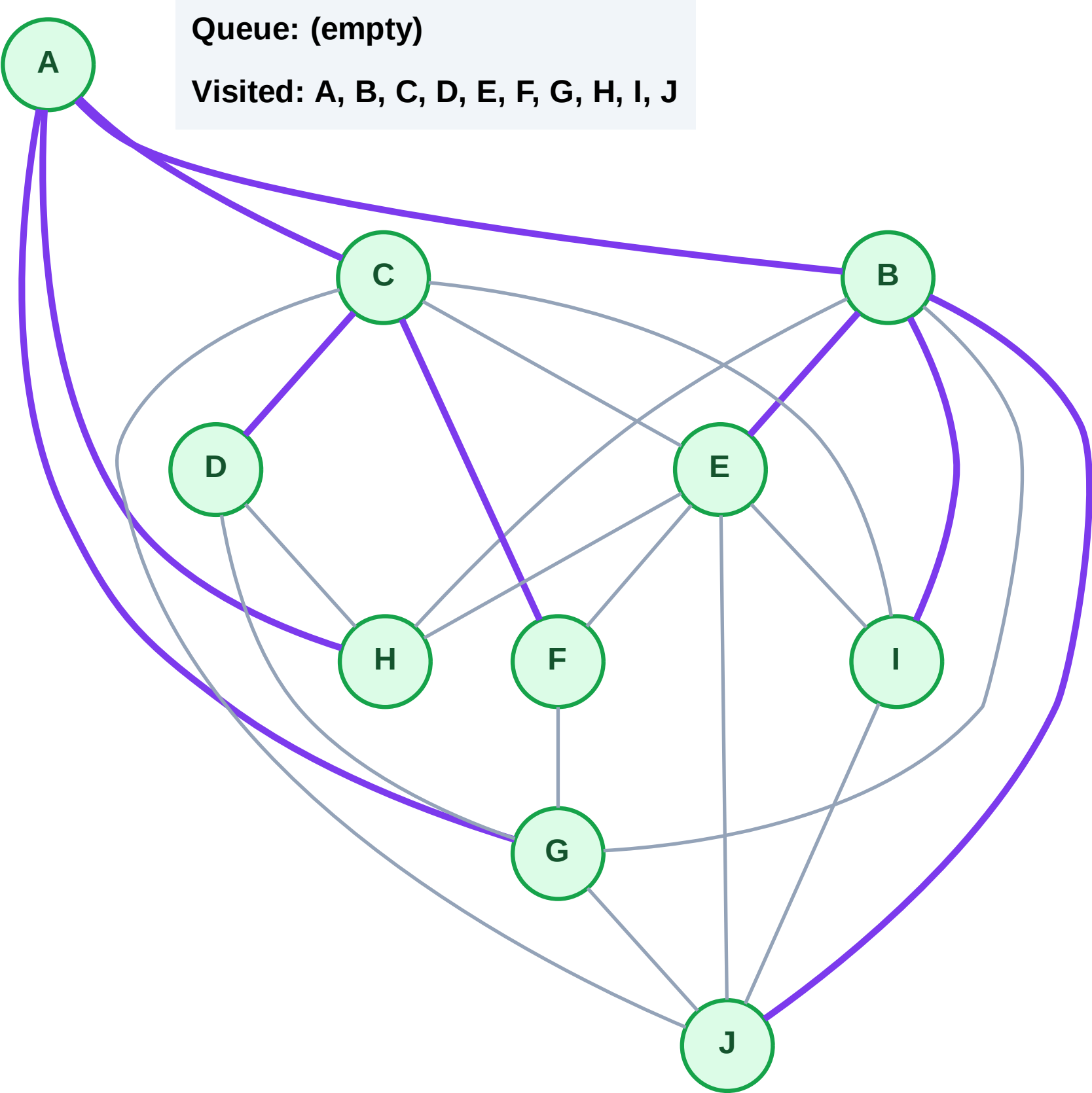
Step 21: No new neighbors from F

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 1: Start at A

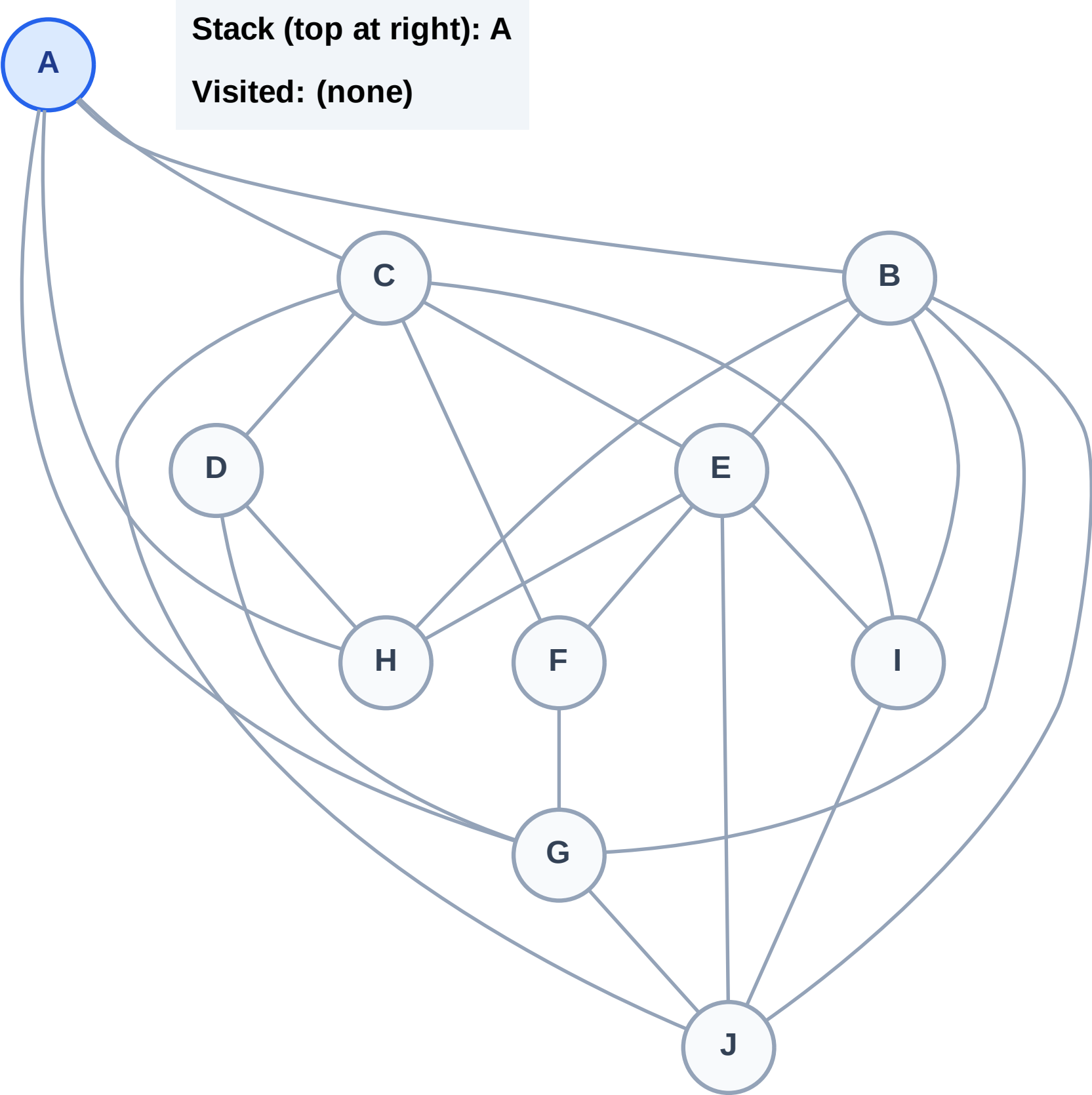
Legend

Complete

Processing

Discovered

Unseen



# DFS Traversal

Step 2: Process node A

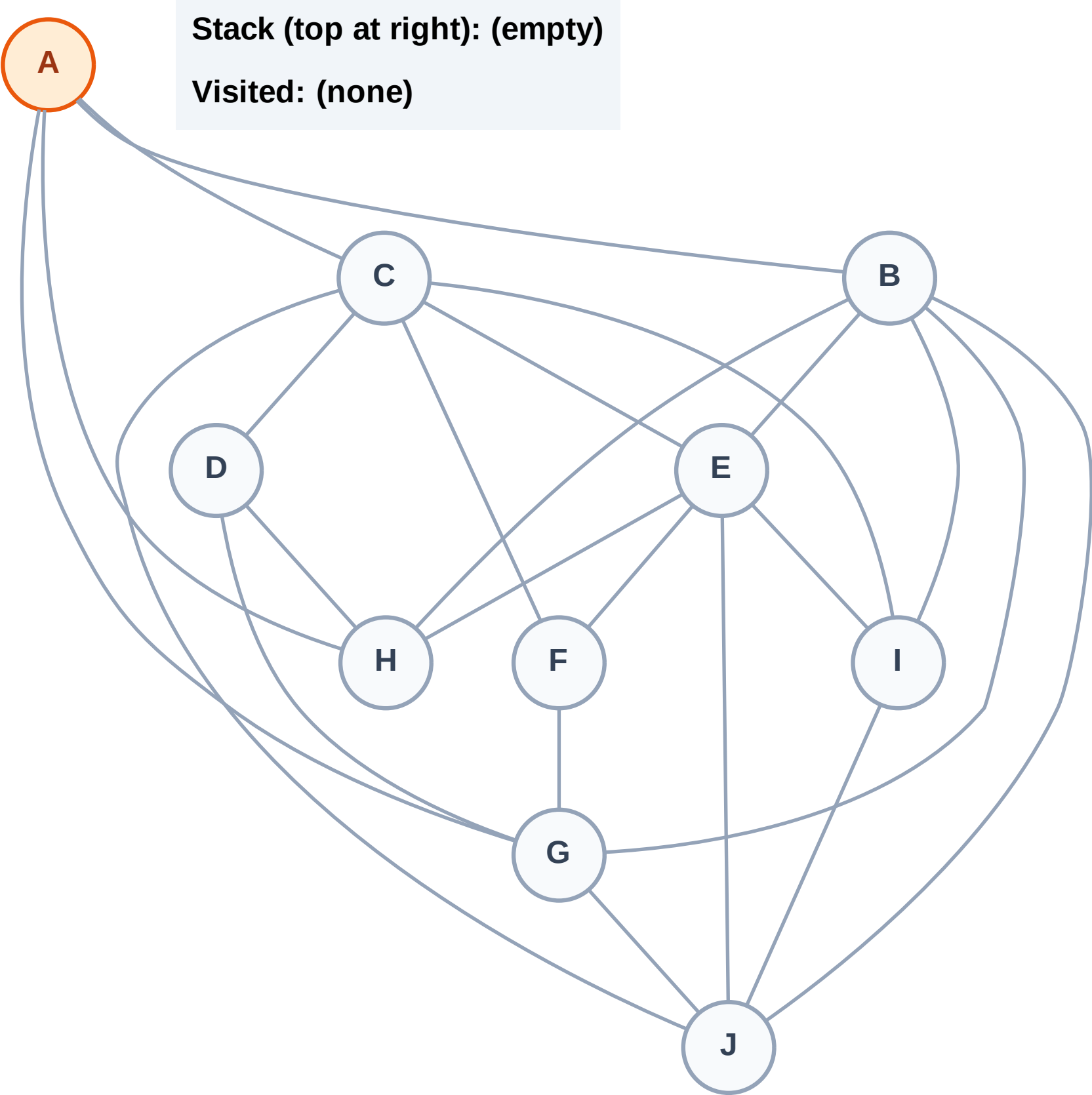
Legend

Complete

Processing

Discovered

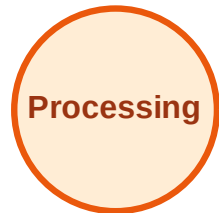
Unseen



# DFS Traversal

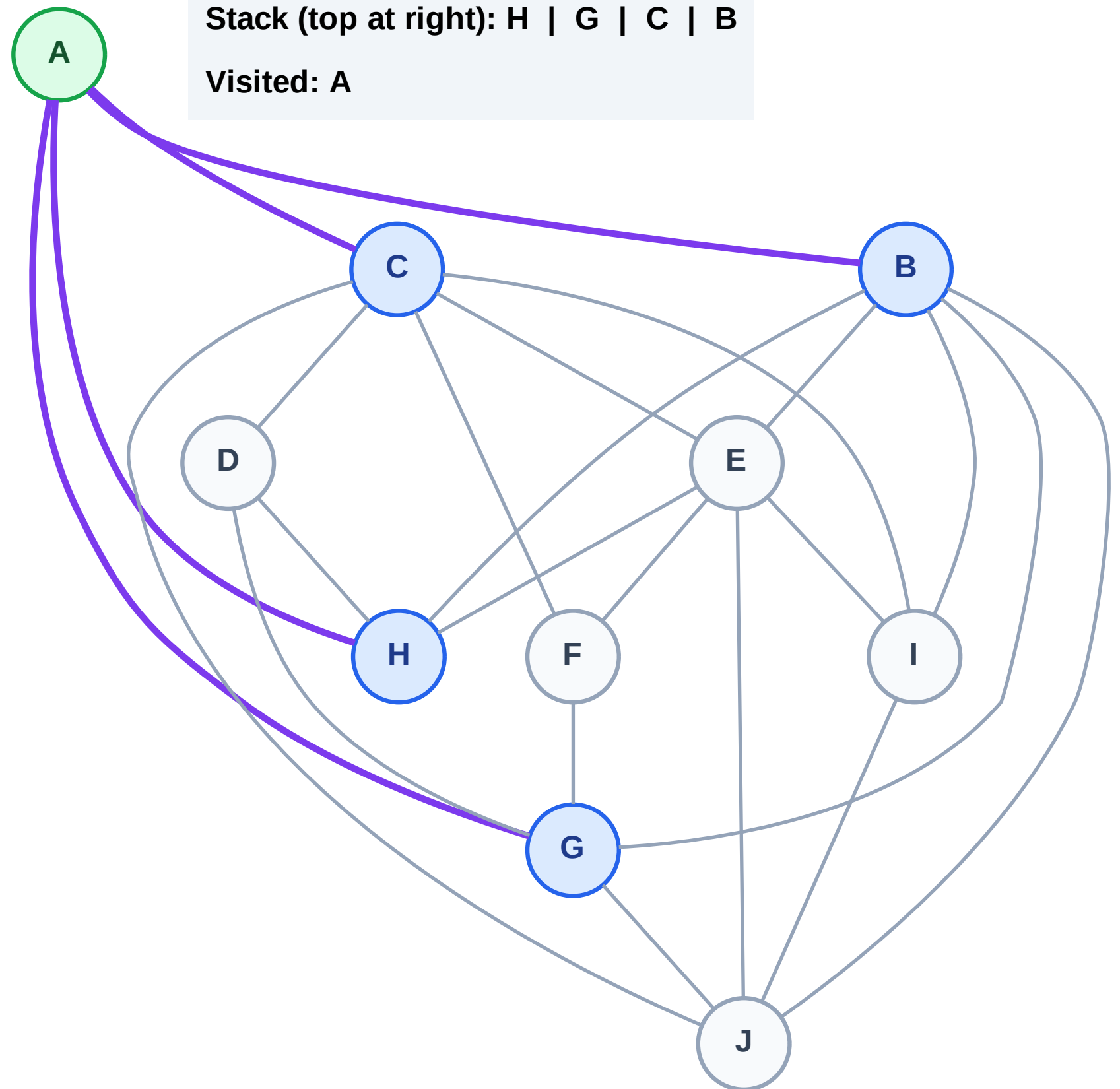
Step 3: Discover H, G, C, B from A

## Legend



Stack (top at right): H | G | C | B

Visited: A

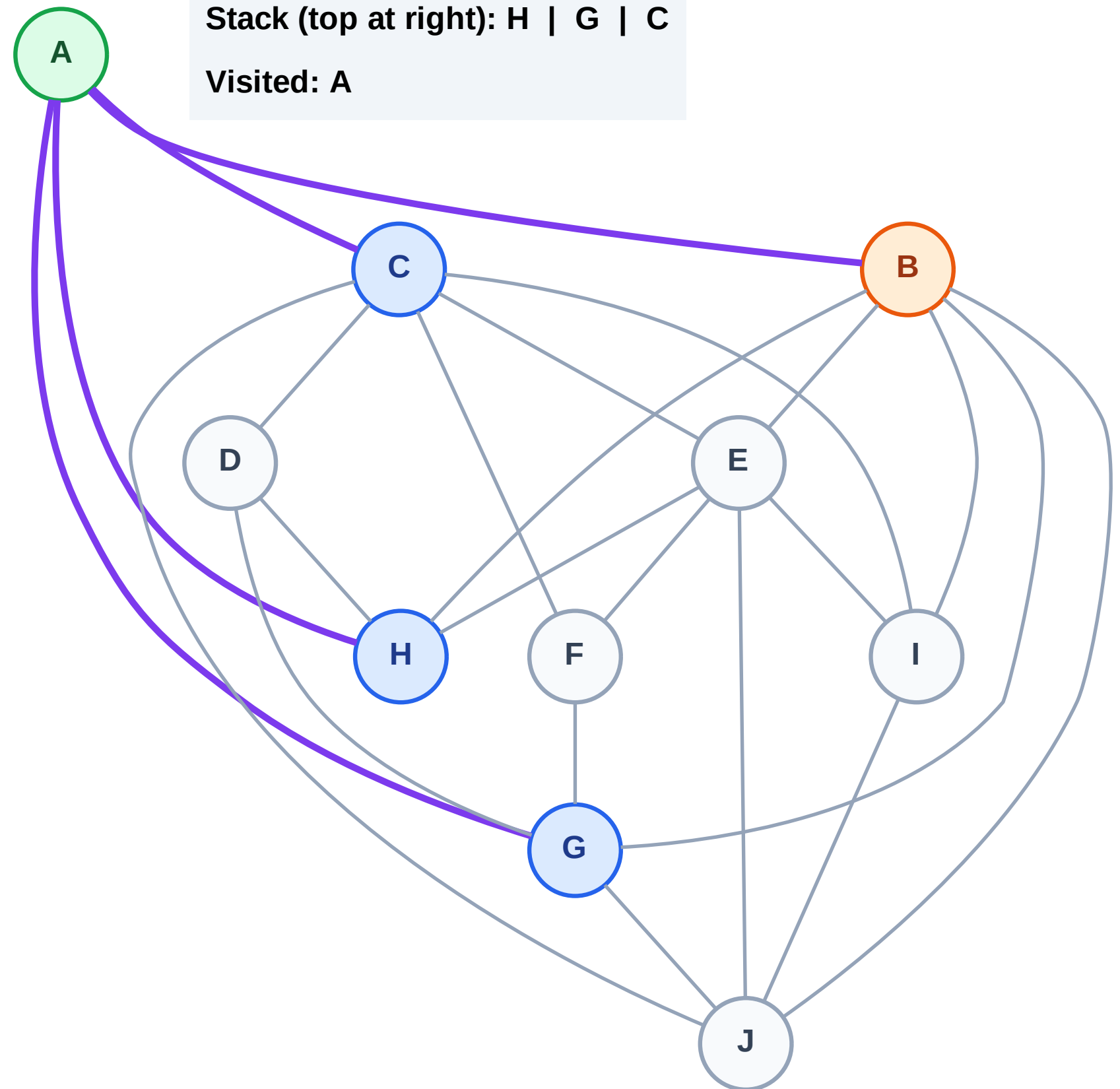
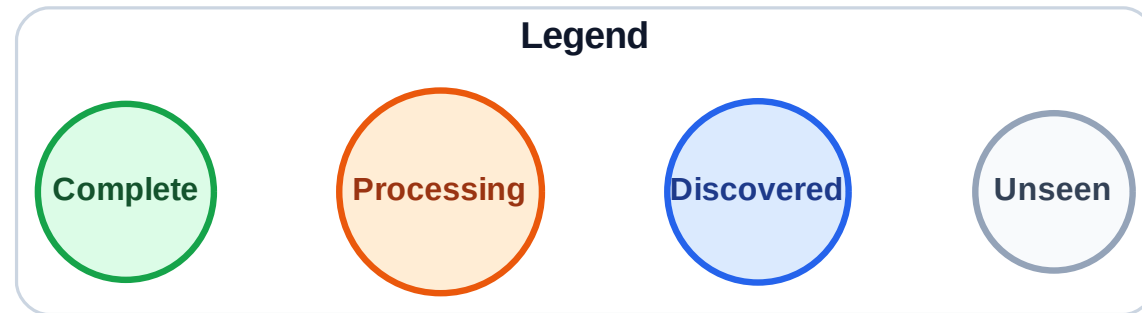


# DFS Traversal

Step 4: Process node B

Stack (top at right): H | G | C

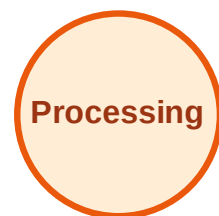
Visited: A



# DFS Traversal

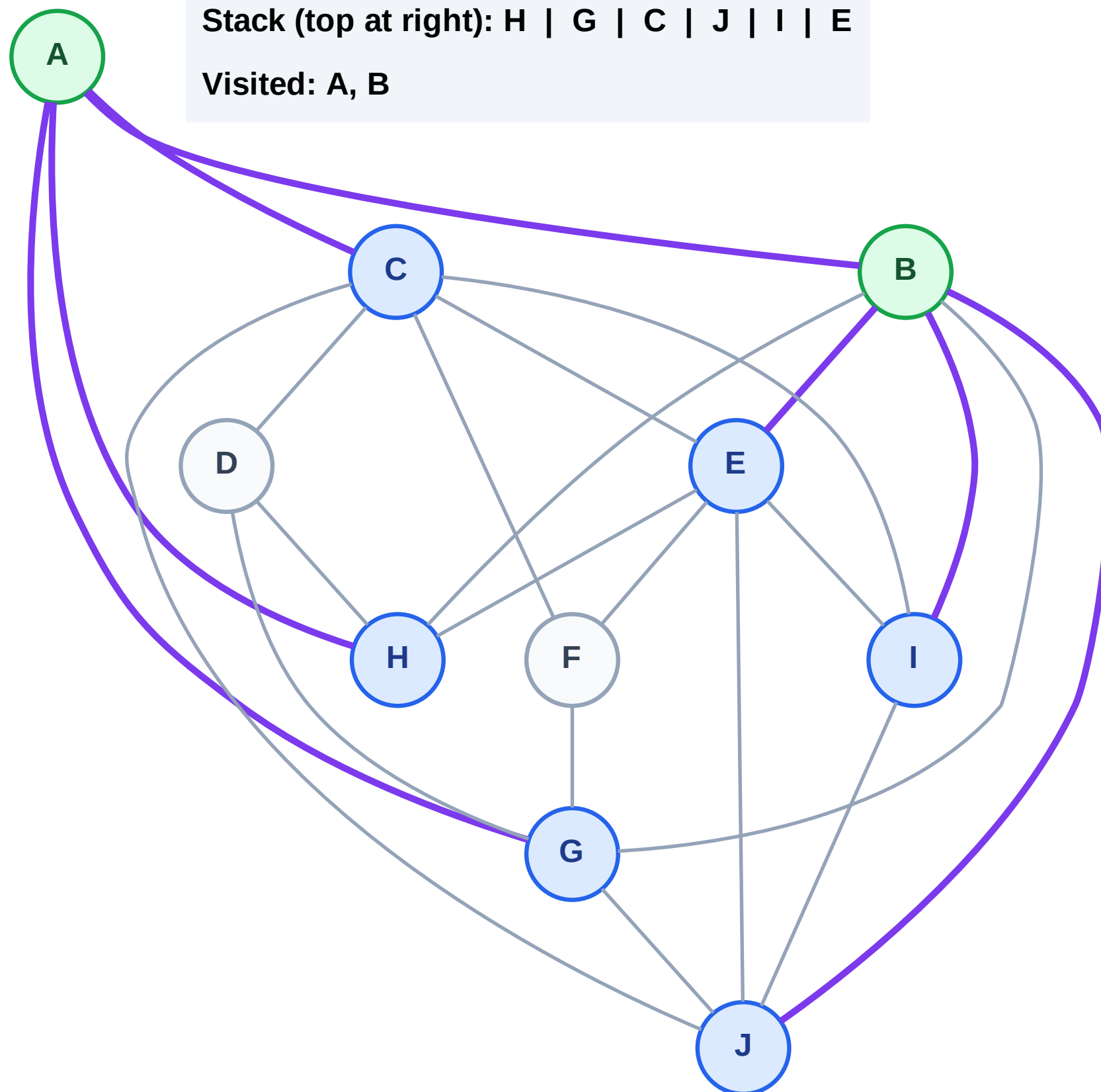
Step 5: Discover J, I, E from B

## Legend



Stack (top at right): H | G | C | J | I | E

Visited: A, B



# DFS Traversal

Step 6: Process node E

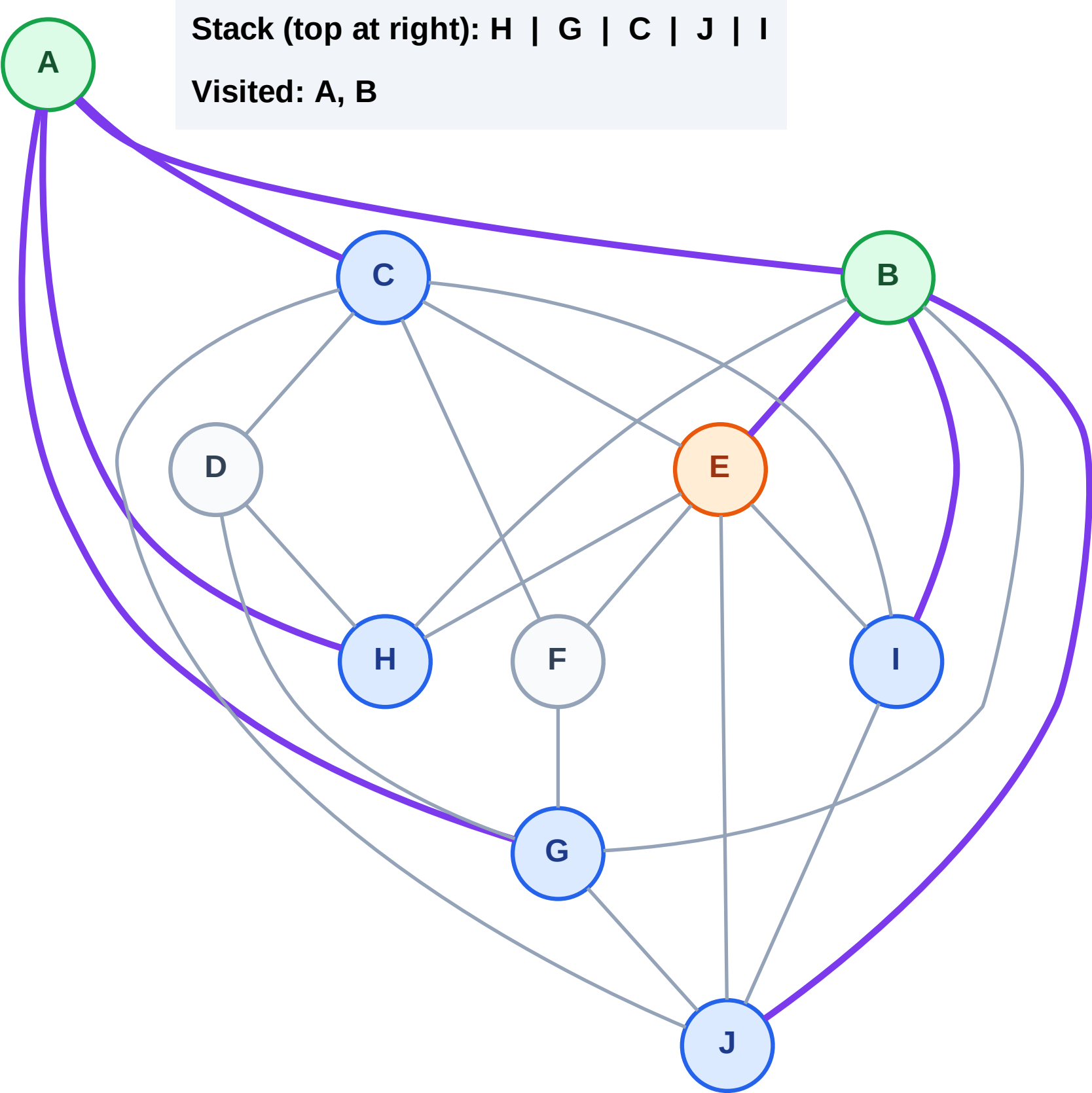
Legend

Complete

Processing

Discovered

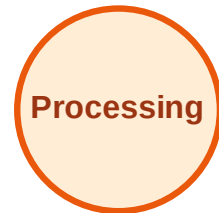
Unseen



# DFS Traversal

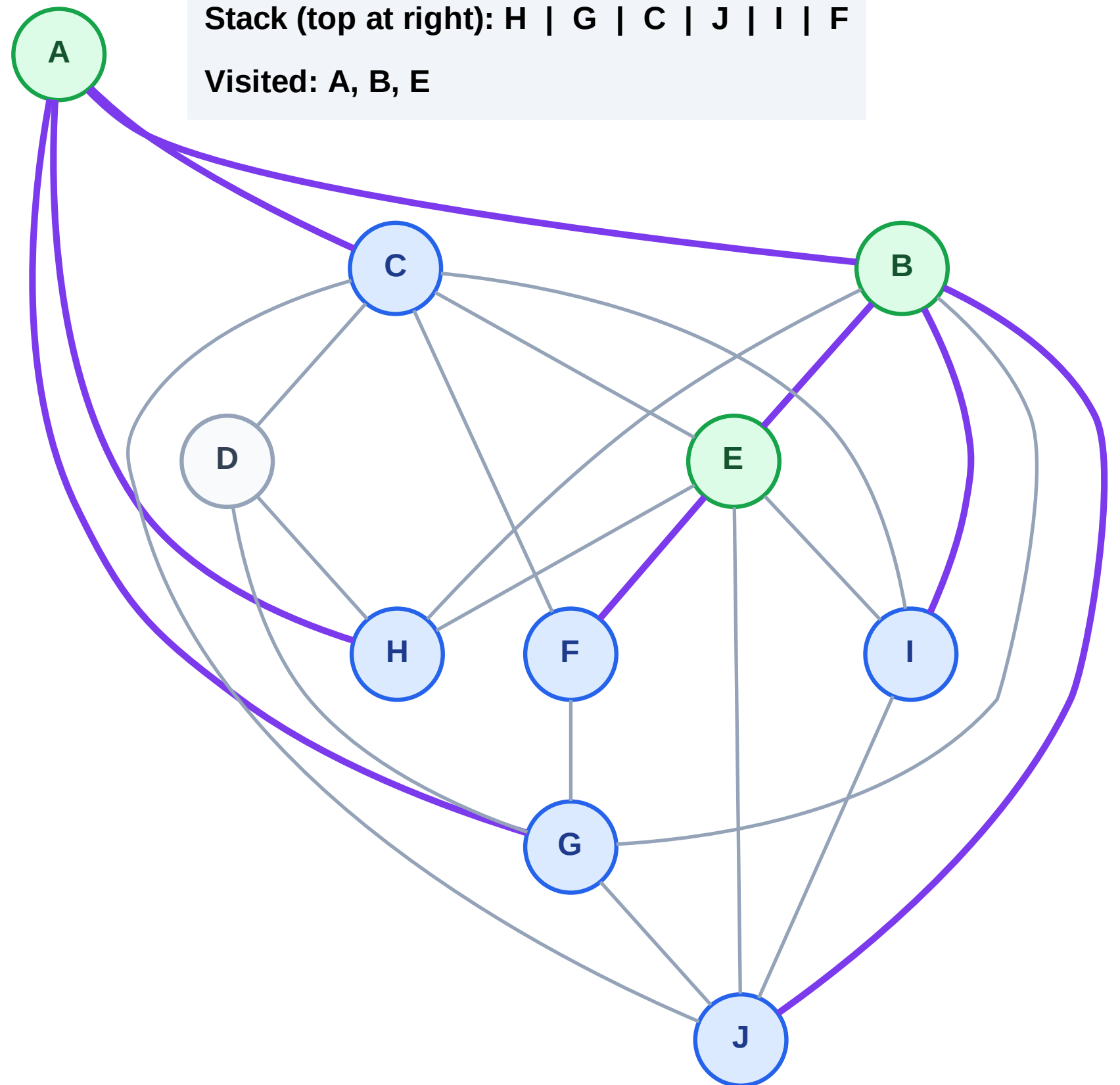
Step 7: Discover F from E

## Legend



Stack (top at right): H | G | C | J | I | F

Visited: A, B, E



# DFS Traversal

Step 8: Process node F

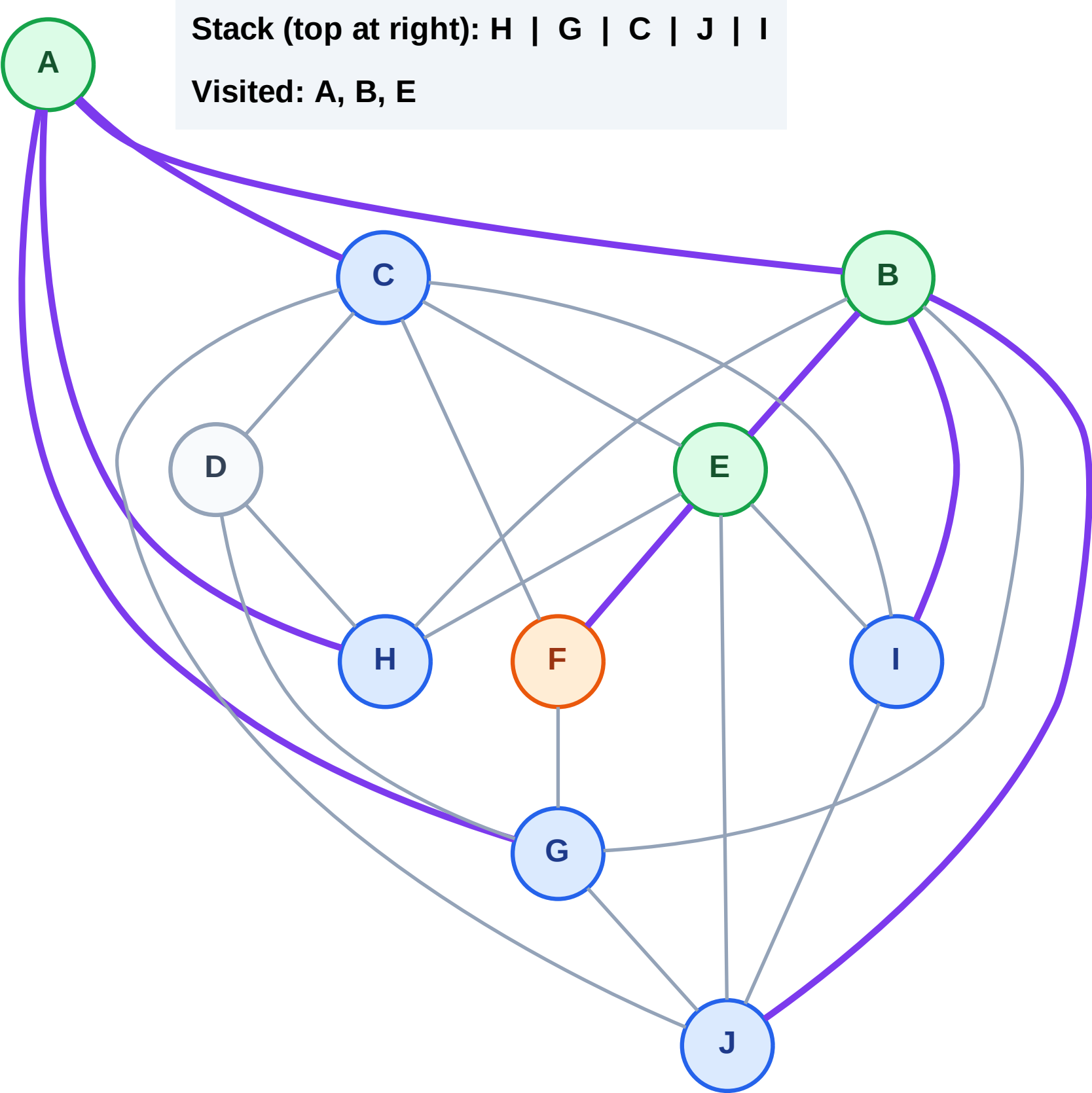
Legend

Complete

Processing

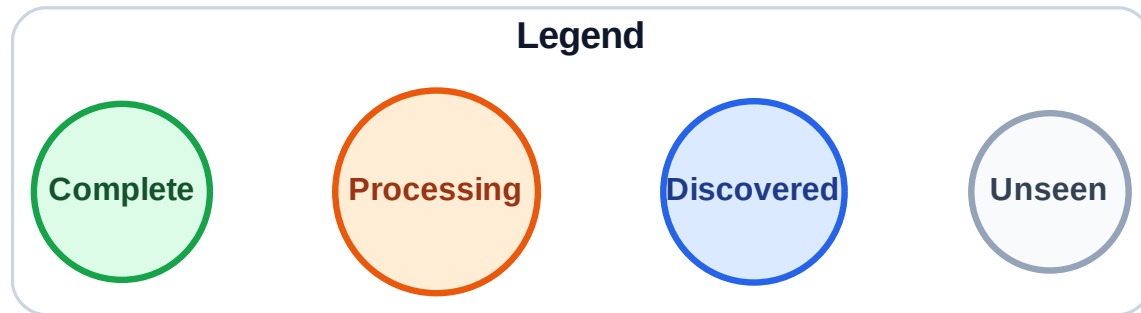
Discovered

Unseen



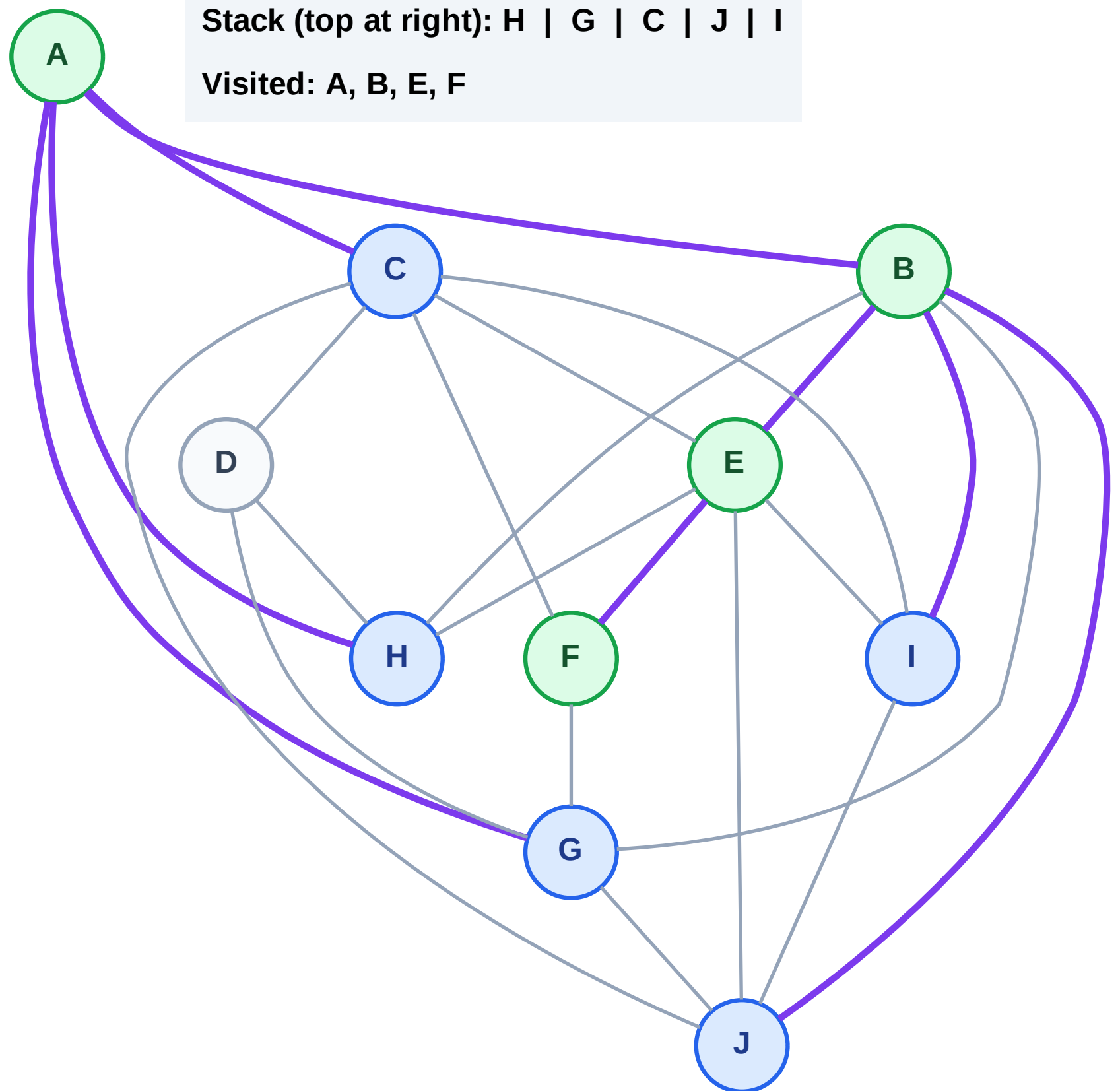
Step 9: No new neighbors from F

Step 9: No new neighbors from F



**Stack (top at right):** H | G | C | J | I

**Visited: A, B, E, F**



# DFS Traversal

Step 10: Process node I

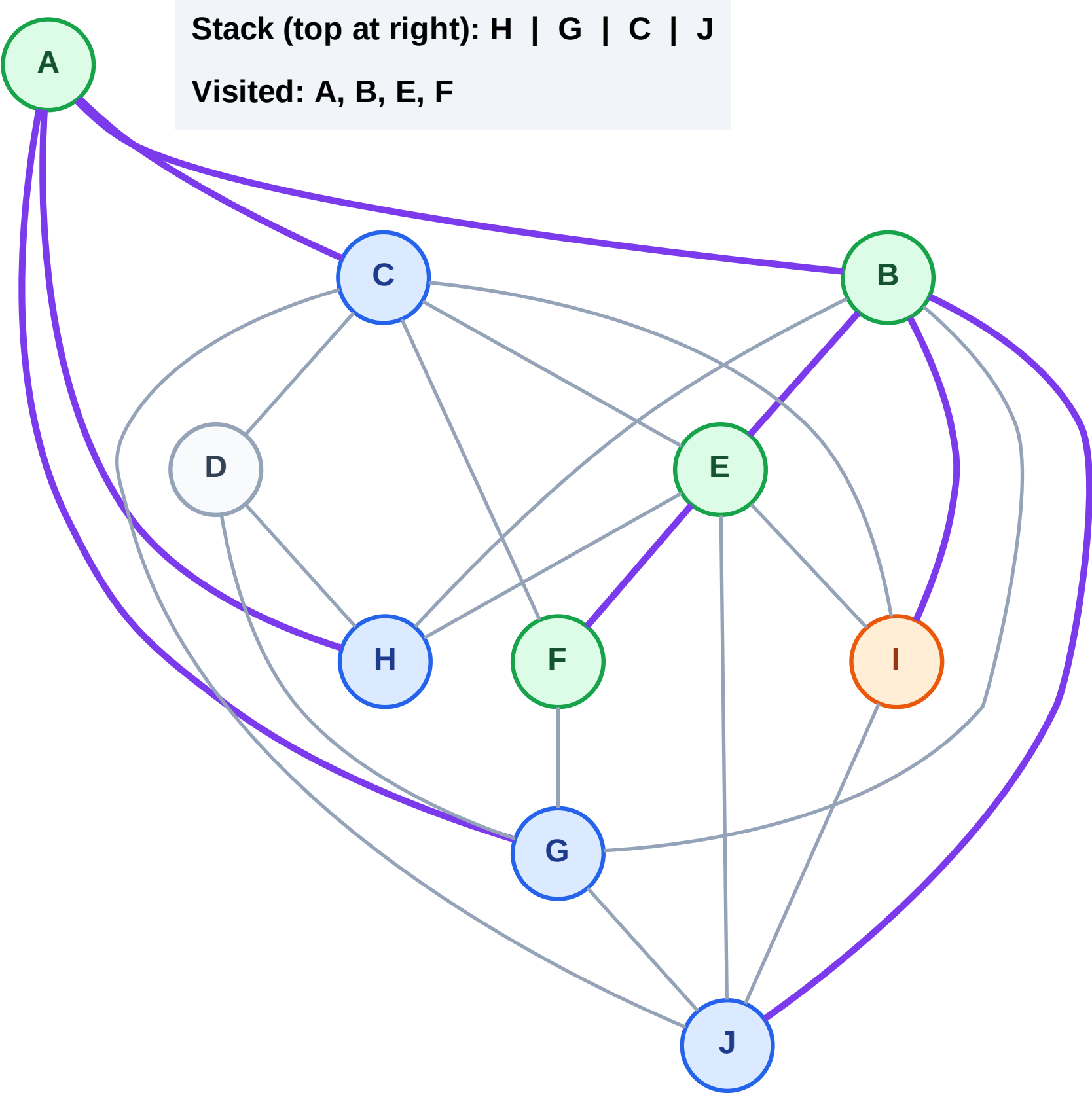
Legend

Complete

Processing

Discovered

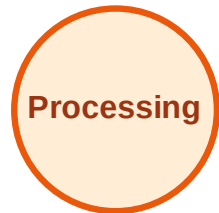
Unseen



# DFS Traversal

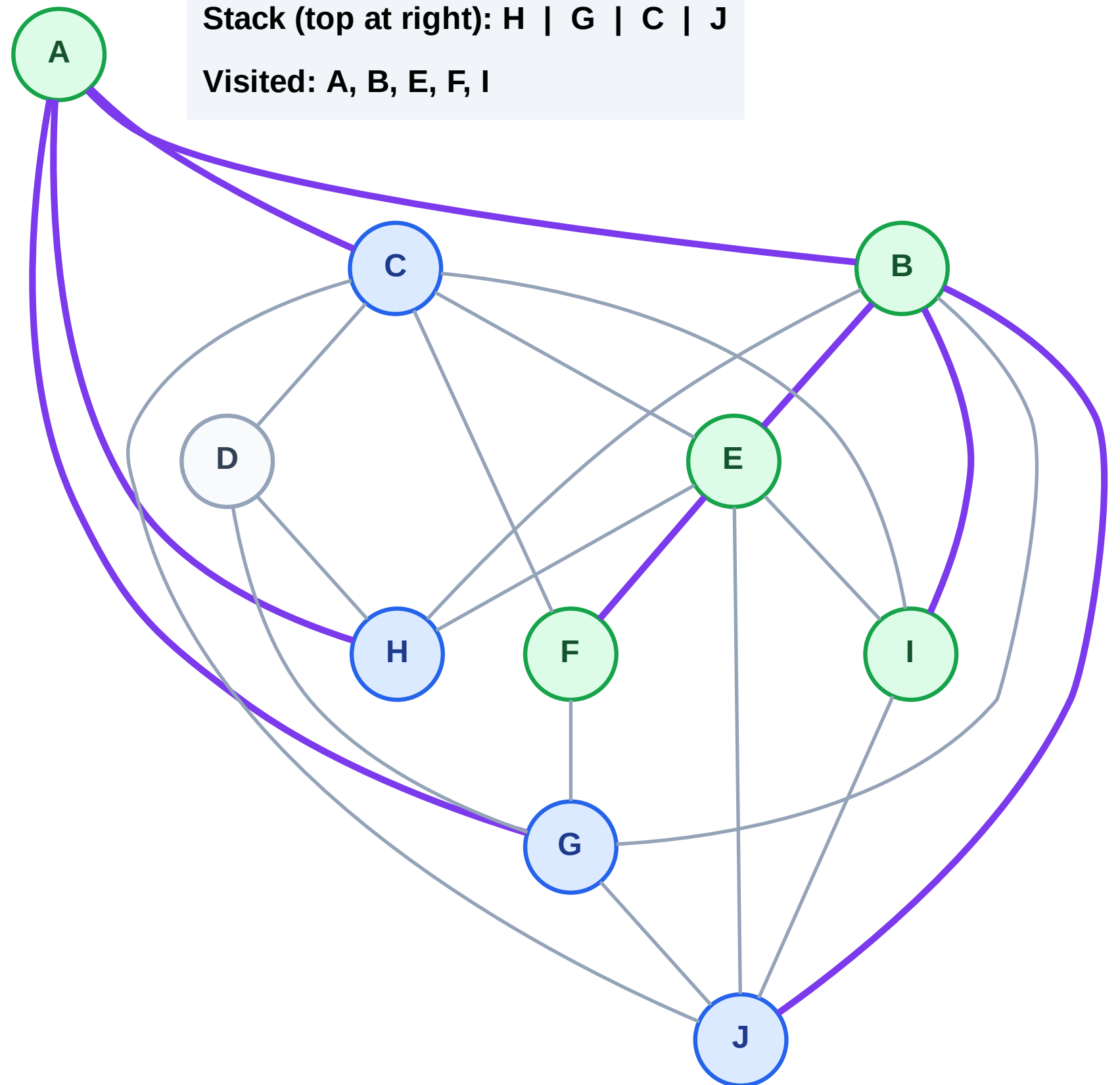
Step 11: No new neighbors from I

## Legend



Stack (top at right): H | G | C | J

Visited: A, B, E, F, I



# DFS Traversal

Step 12: Process node J

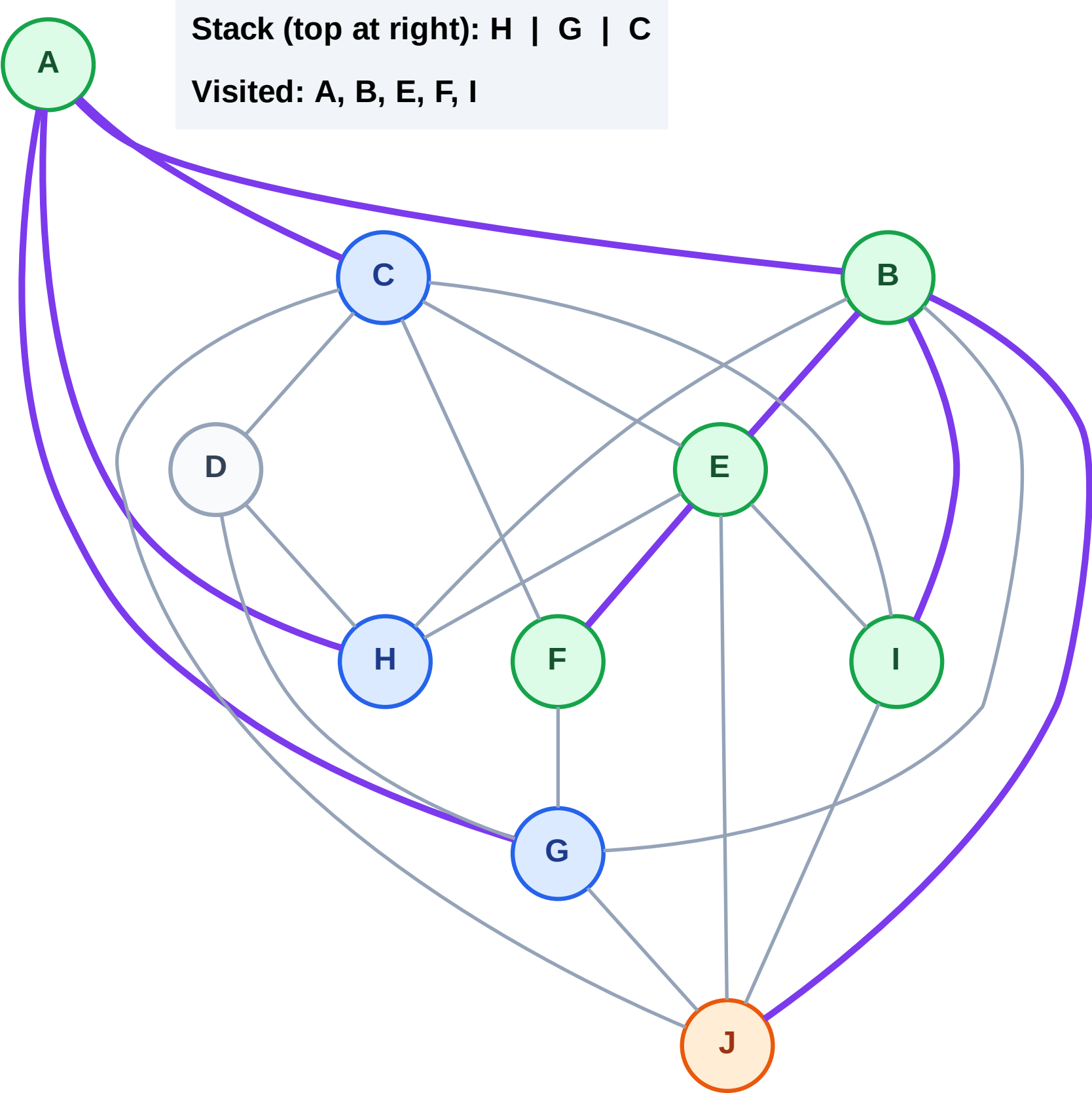
Legend

Complete

Processing

Discovered

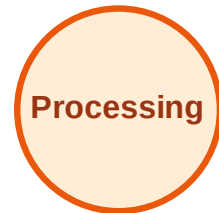
Unseen



# DFS Traversal

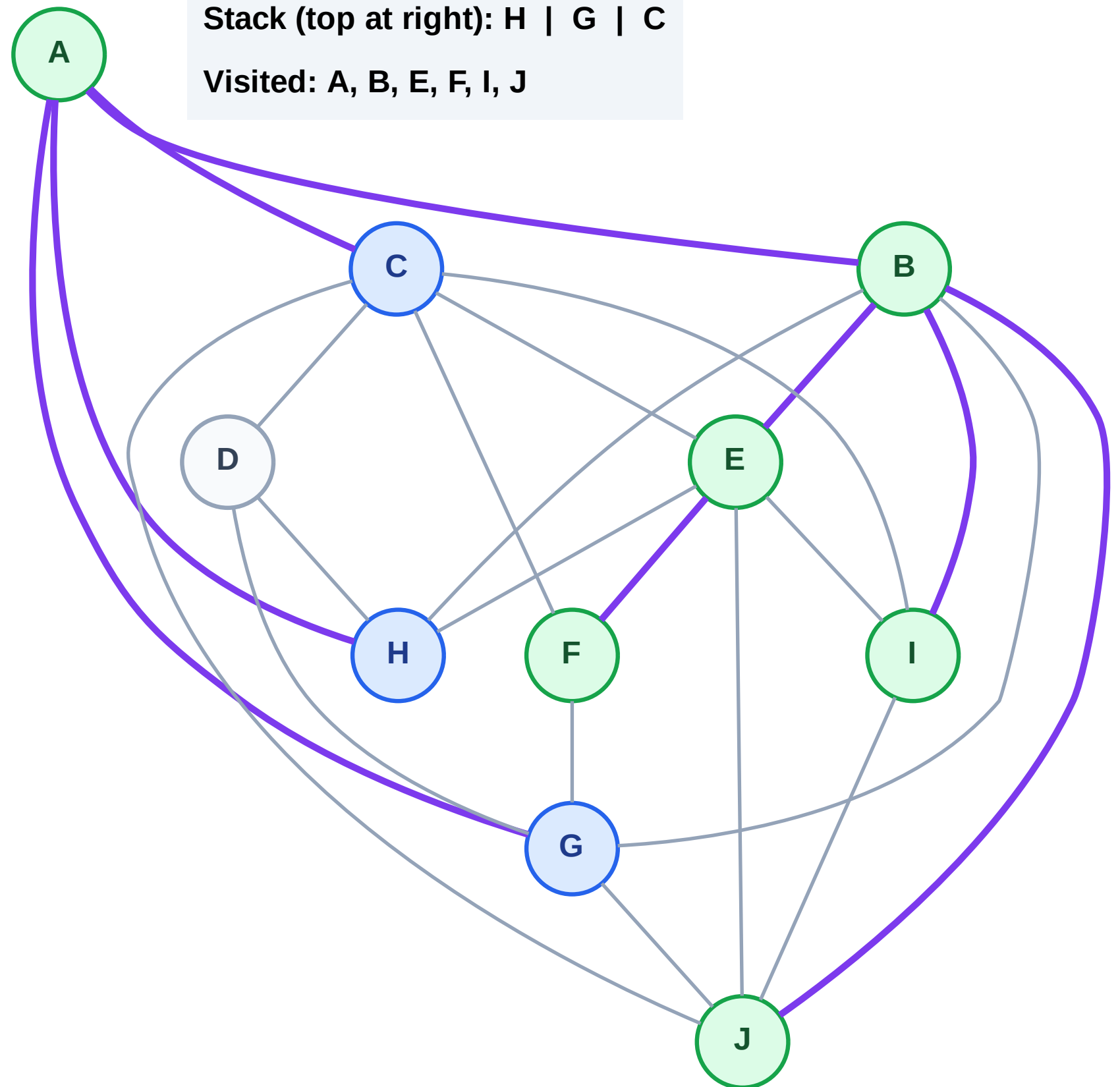
Step 13: No new neighbors from J

## Legend



Stack (top at right): H | G | C

Visited: A, B, E, F, I, J



# DFS Traversal

Step 14: Process node C

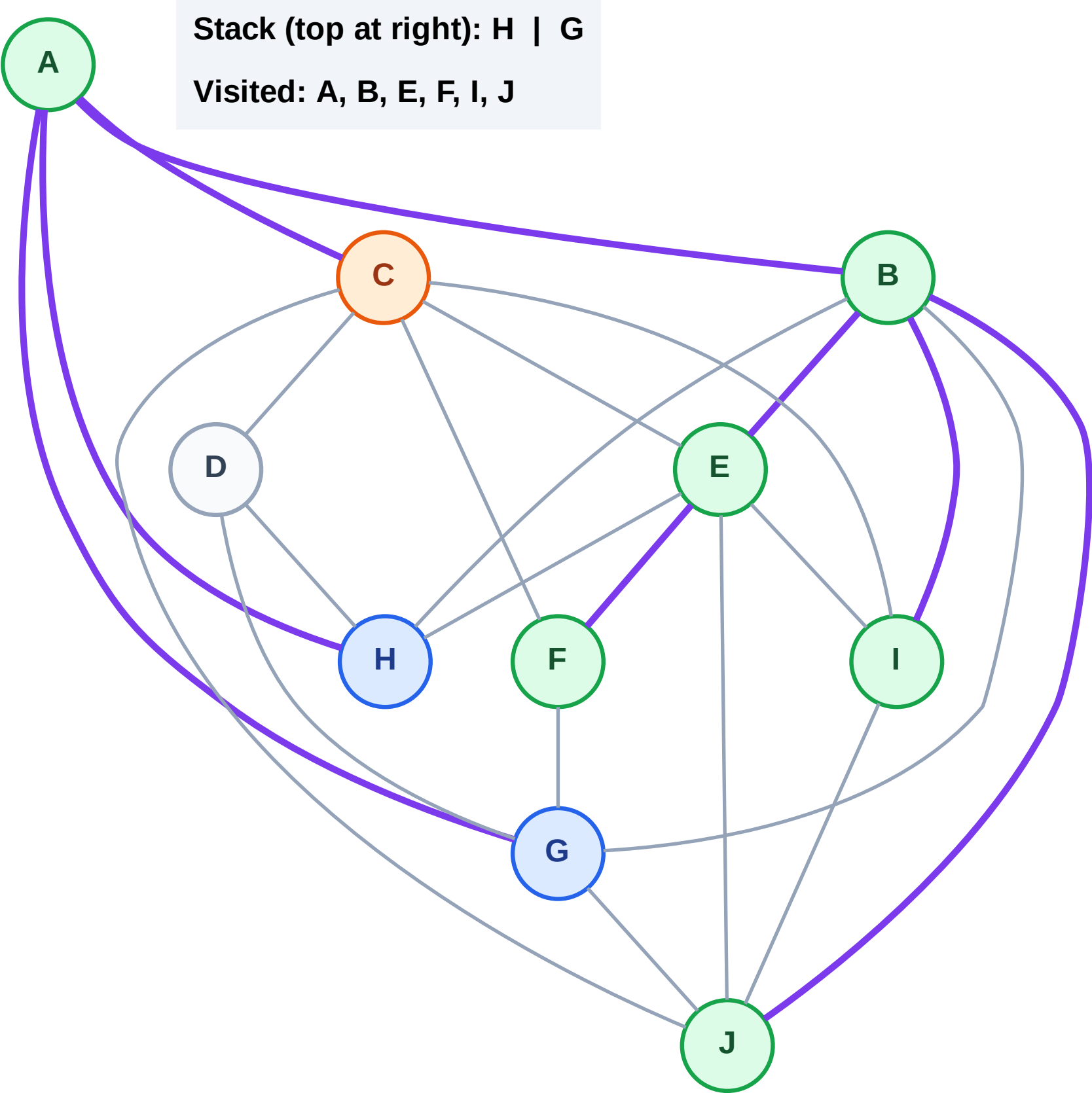
Legend

Complete

Processing

Discovered

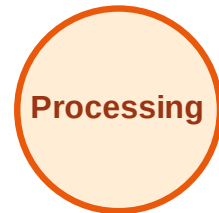
Unseen



# DFS Traversal

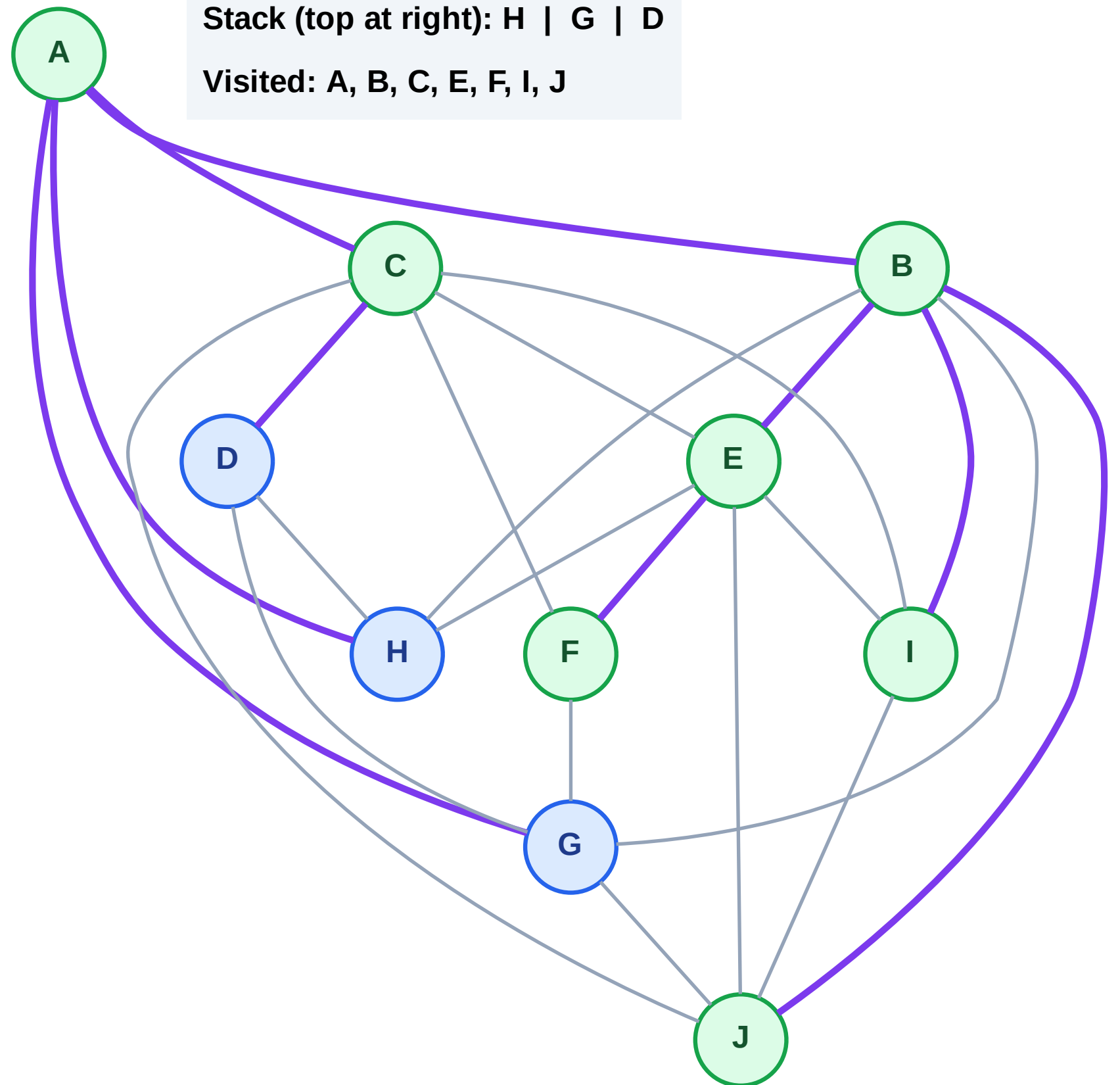
Step 15: Discover D from C

## Legend



Stack (top at right): H | G | D

Visited: A, B, C, E, F, I, J



# DFS Traversal

Step 16: Process node D

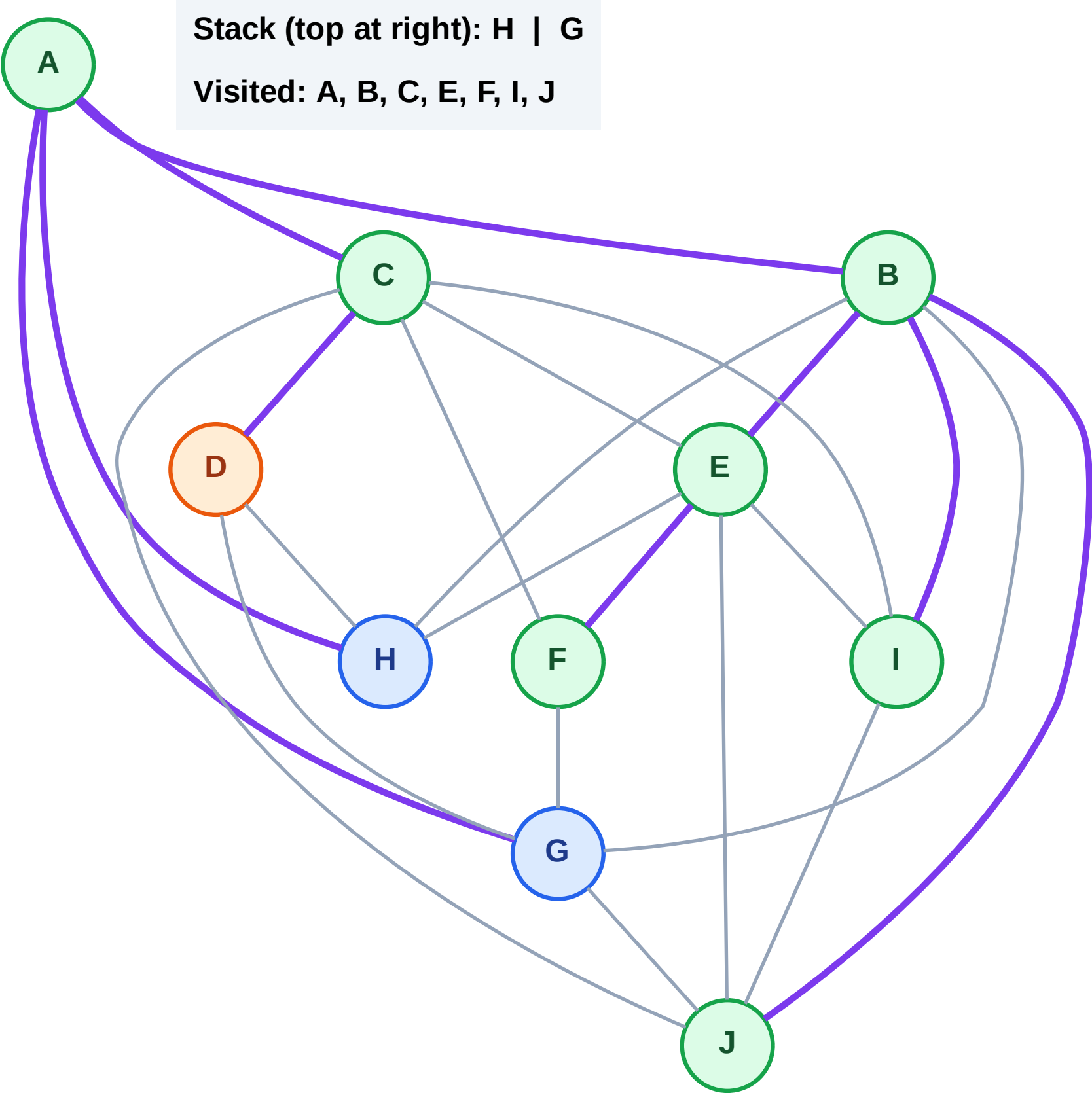
Legend

Complete

Processing

Discovered

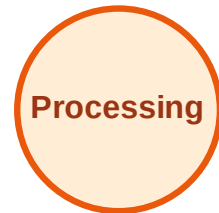
Unseen



# DFS Traversal

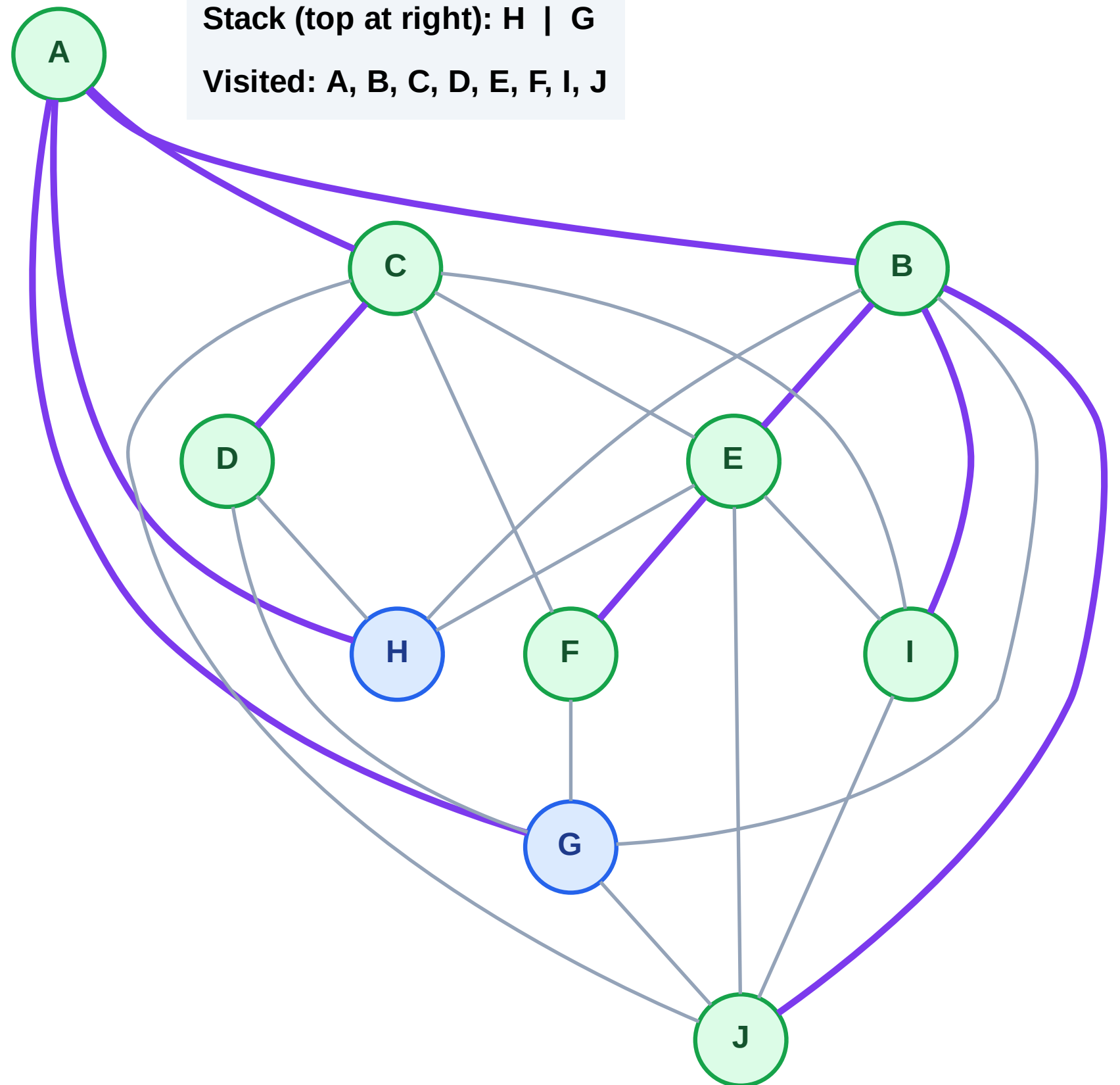
Step 17: No new neighbors from D

## Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



# DFS Traversal

Step 18: Process node G

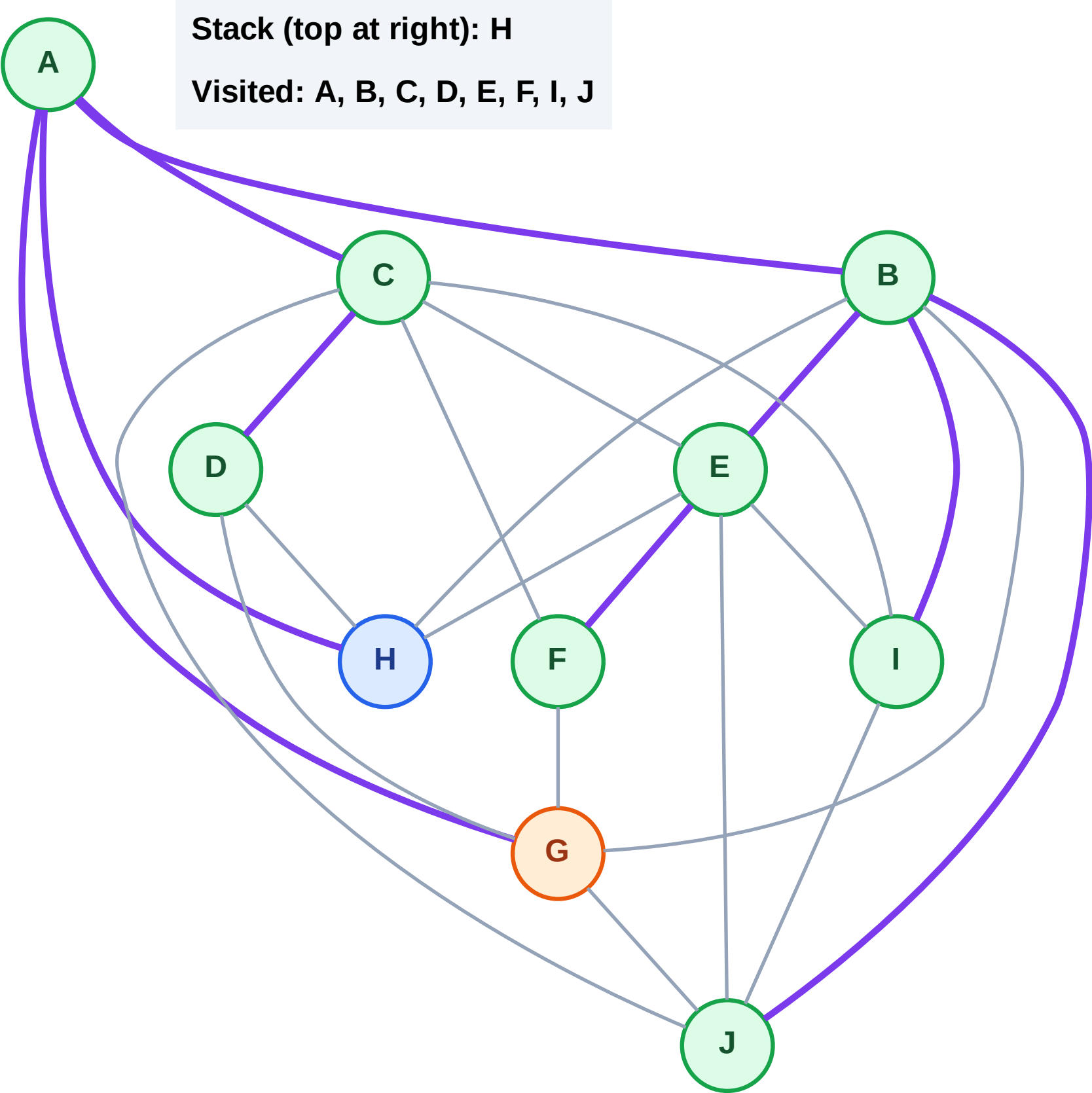
Legend

Complete

Processing

Discovered

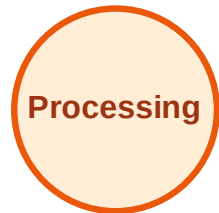
Unseen



# DFS Traversal

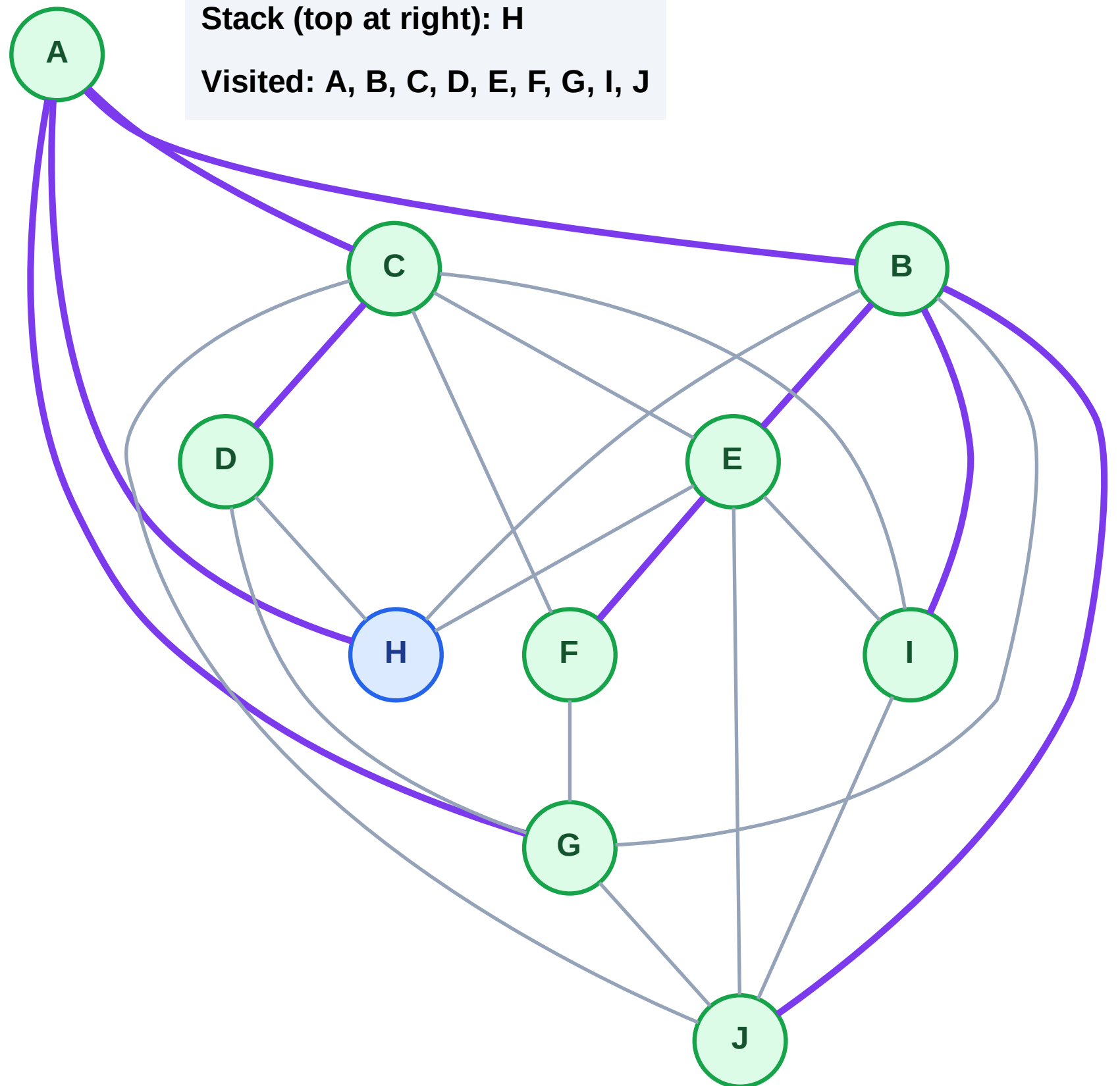
Step 19: No new neighbors from G

## Legend



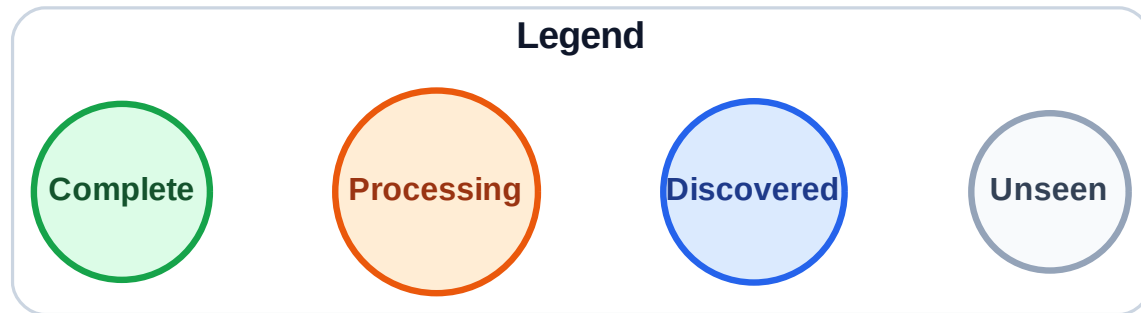
Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



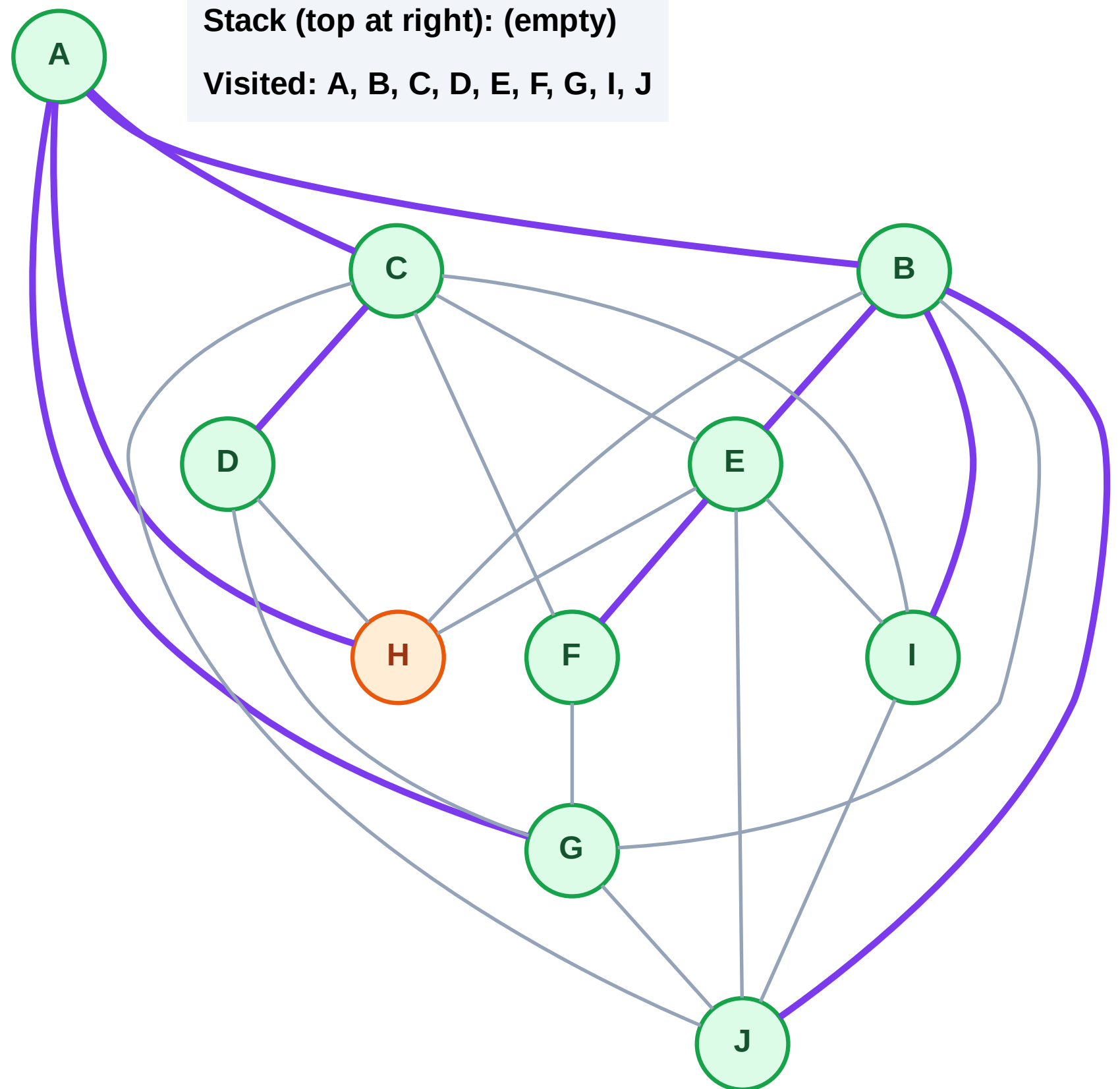
### Step 20: Process node H

### Step 20: Process node H



**Stack (top at right): (empty)**

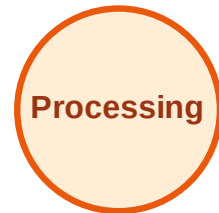
**Visited: A, B, C, D, E, F, G, I, J**



# DFS Traversal

Step 21: No new neighbors from H

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

